Omni Pact White Paper

Web3.0 Global Universal Trust Infrastructure

V1.0.2

catalogue

1. Project Core Overview

- 1.1 Trust Dilemma of Centralized Trading
- 1.2 Opportunities of Trust Revolution in Web3.0 Era
- 1.3 Limitations of Existing On-chain Trading Solutions
- 1.4 Project Mission, Vision and Core Positioning
- 1.5 Core Value Proposition and Core Pain Points Addressed
- 1.6 Target Users and Application Scenario Boundaries
- 1.7 White Paper Reading Guide

2. Core Technology Architecture

- 2.1 Overall Architecture Design
- 2.2 Core components at the protocol layer
- 2.3 Service Layer Support Components
- 2.4 Application Layer Interaction Components
- 2.5 Technical Security and Audit Mechanism

3. Core working mechanisms

- 3.1 Standard secured transaction process (creation-lock-up-performance-confirmation)
- 3.2 Full Dispute Resolution Process (Initiation, Evidence Submission, Arbitration, Enforcement)
- 3.3 Cross-chain Guarantee Transaction Mechanism
- 3.4 Batch Transaction Guarantee Mechanism (e.g., Airdrops)
- 3.5 Credit Score Calculation and Application Mechanism
- 3.6 Smart Contract Trigger Conditions and Execution Logic

4. Product Services and Application Scenarios

- 4.1 Panoramic View of Contract Guarantee Categories
- 4.2 Core Functional Features
- 4.3 Product Iteration Roadmap

5. Token Economy Model

- 5.1 Basic Information of the Token
- 5.2 Core Uses of Tokens
- 5.3 Token Allocation Scheme
- 5.4 Detailed Rules for the Incentive Gradient
- 5.5 Token Destruction Mechanism
- 5.6 Analysis of the Sustainability of the Economic Model

6. Business and Operations Planning

- 6.1 Business Model
- 6.2 Market Expansion Strategies
- 6.3 User Growth and Retention
- 6.4 Community Operations and Governance
- 6.5 Operational Milestones

7. Ecological Development Planning

- 7.1 Short-term ecological goals (1-2 years)
- 7.2 Medium-term Ecological Layout (3-5 years)
- 7.3 Long-term ecological vision (5 years+)
- 7.4 Developer Ecosystem Support Program
- 7.5 Recruitment of Ecological Partners and Cooperation Models

8. Compliance Framework

- 8.1 Analysis of the current status of global regulatory compliance
- 8.2 Core Compliance Principles and Strategies

- 8.3 Data Security and Privacy Compliance
- 8.4 Compliance with asset transaction regulations
- 8.5 Cross-border transaction compliance solutions
- 8.6 Compliance Iteration and Adaptation Mechanism

9. Risk Prevention and Warning

- 9.1 Risk Prevention and Control System
- 9.2 Emergency Response Plan
- 9.3 Potential Risk Warning

1. Project Core Overview

1.1 The Trust Dilemma of Centralized Transactions

Over the past three decades of internet commercialization, the global digital trade boom was built on a single assumption: trust between strangers must be established through a powerful centralized intermediary (Trusted Third Party).

From e-commerce giants like Amazon and eBay to gig economy platforms such as Upwork and Fiverr, and traditional payment gateways including PayPal and Swift, these centralized entities have built the infrastructure of the digital economy. By providing credit guarantees, escrow services, and dispute resolution mechanisms, they mitigated early transactional friction. However, as the global digital economy evolves into deeper and broader dimensions, the marginal benefits of this "agency trust model" are diminishing. Its structural flaws—what we call "the tyranny of intermediaries" —are increasingly becoming the biggest obstacle to the free flow of value.

The current centralized trading system has trapped global commerce in four major structural challenges:

1. Value Snatching: High "Trust Tax" and Economic Friction

Centralized platforms are not public infrastructure but commercial entities that prioritize shareholder interests. To sustain their massive audit teams, compliance costs, and monopoly profits, these platforms impose steep fees on ecosystem participants.

- Commission-based exploitation: Major platforms typically charge 10% to 30% of transaction volumes as service fees. For cross-border payments, an additional 3% to 5% is deducted as exchange rate spreads and settlement fees. This hefty 'trust tax' severely erodes producers' profit margins and discourages micro-payments and high-frequency trading.
- Rent-seeking mechanism: The platform leverages its monopoly position to force merchants into secondary payments through tactics like paid ranking and traffic distribution control, thereby exacerbating the deterioration of the business environment.

2. Sovereignty Loss: Single Point of Failure and Asset Control

In a centralized architecture, users do not actually own their accounts, funds, or data. The platform holds absolute control, creating a significant Single Point of Failure (SPF) risk.

- Asset freeze: The platform may freeze merchants 'funds or suspend accounts at any time without requiring concrete evidence, driven by black-box risk control algorithms or geopolitical pressures. This unpredictable' liquidity sudden death' risk has become a nightmare for countless small and medium-sized enterprises.
- Data Cage: Users 'transaction history, social connections, and reputation ratings are locked in the platform's proprietary database. They cannot transfer their decade-long reputation from Platform A to Platform B. This vendor lock-in deprives users of data sovereignty.

3. The black box of adjudication: opaque dispute resolution mechanisms

When disputes arise (e.g., goods not matching the description or services failing to meet standards), the arbitration mechanism of centralized platforms often proves inefficient and biased.

- Algorithmic tyranny: To cut operational costs, a flood of disputes are handled by rigid automated scripts that fail to grasp complex business contexts.
- Systematic bias in rulings: To maintain buyer engagement or avoid chargeback risks, platforms often favor buyers in arbitration, leaving sellers vulnerable to free-riding or fraudulent refunds, with complaint channels being severely limited or even shut down.

4. Global fragmentation: The wall of financial inclusion

The existing trust system is highly dependent on the traditional banking financial network, which leads to the fragmentation of the global market.

- Regional discrimination: Billions of people worldwide (mainly in developing countries) still lack access to mainstream platform account services due to compliance costs or sanctions.
- Inefficient settlement: Due to outdated banking clearing systems, cross-border transaction funds often take T+3 or even T+7 days to be credited, severely hampering capital turnover efficiency.

The current system forces us to choose between "paying a high price for centralized trust" and "taking significant risks for peer-to-peer freedom." This binary dilemma stands as the greatest obstacle to creating a truly open, efficient, and inclusive global value internet. OmniPact was born from a firm conviction: trust should not be a commodity to be sold, but a public infrastructure safeguarded by mathematics and code. Our mission is to build a new paradigm that breaks free from this dilemma.

1.2 The Missing Link in Web3.0 Ecosystem: The Discontinuity Between Assets and Delivery

While Web2.0 grappled with the issue of 'excessive power concentration,' Web3.0 now confronts the core challenge of 'the absence of a business and trust infrastructure.'

Blockchain technology, through asymmetric encryption and distributed ledgers, has perfectly resolved the decentralized issuance and circulation of 'currency,' creating trillions of dollars in on-chain assets. Yet when attempting to exchange these assets for real-world value (goods, services, data, or equity), we find ourselves in a 'trust vacuum' filled with technological gaps and game-theoretic risks.

The Web3.0 ecosystem currently faces three structural deficiencies that prevent its evolution from a 'financial speculation tool' to a 'global commercial settlement layer'.

1. The Limitations of "Atomic Exchange" and the Risk of "Send-and-Forget"

Current smart contracts excel at handling 'atomic swaps' ─instant exchanges between Tokens A and B, as seen on Uniswap. However, in scenarios involving delayed or off-chain delivery, blockchain's' immutability' becomes a double-edged sword.

- Irreversible panic: Once users click 'send', funds are immediately withdrawn and cannot be recovered. If the seller defaults, flees, or delivers substandard products, buyers have no recourse (no customer service, no refund button).
- Dark Forest Principle: The absence of safeguards has turned on-chain OTC trading, service procurement, and P2P transactions into hotspots for fraud, leaving most users with no choice but to rely on centralized exchanges or guarantors for matching─contrary to

the core tenets of decentralization.

2. The Unresolvable "Prophecy Machine Dilemma" in Business Scenarios

Smart contracts are essentially closed-off 'autonomous systems' that can seamlessly access on-chain account balances, yet remain completely unaware of the off-chain world (Real World).

- Information gaps: The contract cannot verify whether the limited-edition sneakers have been delivered via FedEx, if the outsourcing code has passed testing, or if the API interface remains operational.
- The challenge of verification: While oracles like Chainlink have addressed the blockchain integration of financial data (e.g., ETH price updates), the market still lacks mature decentralized verification protocols for non-standardized, highly subjective business activities (e.g., service quality assessments, dispute liability determination). This impedes the conditional release of on-chain funds based on actual outcomes.

3. The Financial Non-Operation and the Decoupling of Real Economy

The current DeFi (decentralized finance) ecosystem exhibits a highly competitive yet superficial 'financial hollowing-out' phenomenon.

- The self-referential cycle: Users stake tokens to borrow more, then purchase additional tokens. While funds flow between protocols, they rarely enter the real economy to generate actual productivity or consumer value.
- The absence of a closed-loop business ecosystem: Due to its inability to resolve real-world disputes and provide transaction guarantees, Web3.0 has yet to support large-scale e-commerce, supply chain finance, or freelance markets. This has led to crypto assets being long criticized as 'Ponzi schemes,' making it difficult for them to gain mainstream commercial acceptance.

The Necessity of the Trust Layer

Web2.0 monopolized trust, Web3.0 abandoned trust, and the true future lies in 'reconstructing trust'.

The market urgently requires a middleware protocol that inherits blockchain's' decentralization and security 'while delivering' flexibility and arbitration capabilities 'for complex real-world transactions. It must bridge the significant gap between' on-chain assets 'and' off-chain delivery'.

This is precisely OmniPact purpose: it is not meant to replace existing blockchains, but to provide a protective layer for this bare financial chain, ensuring secure implementation of commercial transactions.

1.3 Project Mission, Vision and Core Positioning

The OmniPact Protocol was not created to patch the flaws of the old world, but to lay the foundation for a new one.

1. Mission: Making Trust a Public Infrastructure

"To democratize trust by turning it from a privatized commodity into a public utility."

transforming trust from a commodity of privatization to a public utility of decentralization.

Our mission is to leverage blockchain smart contract technology to redefine the fundamental principles of global commercial credit. We aim to dismantle the monopoly of giants on "trust

rights" and establish a decentralized guarantee network that requires no authorization, is tamper-proof, and accessible to all. We strive to enable any two strangers on Earth—whether in New York or Nairobi—to conduct secure and transparent value exchanges with mathematical certainty, without the need for third-party intermediaries.

2. Vision: Building a Value Contract Network for the Internet of Everything

"A borderless, permissionless economy where value flows as freely as information."

(An economy without borders or permits, where value flows as freely as information.)

We envision a future in which:

- All assets can be secured: From over-the-counter (OTC) transactions of crypto assets, to code delivery and service outsourcing, to cross-border transfers of real-world assets (RWA), every commercial activity can be encapsulated and settled through the OmniPact protocol.
- Global credit interoperability: Business reputation is no longer confined to a single platform, but has evolved into on-chain assets owned by users, universally applicable across all OmniPact network-connected platforms worldwide.
- The friction coefficient of business reaches zero: fraud disappears, intermediary fees are eliminated, and the efficiency of global collaboration reaches unprecedented heights.

3. Core Positioning: The Trust and Settlement Layer of Web3.0 Economy

OmniPact transcends being merely a DApp application or a standalone DeFi protocol. Within the Web3.0 technology stack, we occupy a unique and indispensable niche:

Trust Settlement Layer

- Downward (linking on-chain assets): We operate on major public blockchains like Ethereum, BNB Chain, and Solana, leveraging their security features for fund custody. As middleware, we bridge blockchain finance systems with commercial applications.
- Upward (Supporting Business Applications): We provide standardized collateral API and SDK for upper-layer applications including decentralized marketplaces, OTC platforms, freelancer markets, and supply chain finance solutions. Businesses can leverage OmniPact to access premium collateral services without reinventing the wheel.
- Connecting to the Real World: Through Oracles and Decentralized Arbitration Networks (DAN), we extend on-chain logic to off-chain delivery, serving as a bridge between On-chain Assets and Off-chain Reality.

In short, if Chainlink serves as the data layer of Web3 and Uniswap as its transaction layer, then OmniPact acts as the foundational credit layer of Web3.

1.4 Core Value Proposition and Core Pain Points (Core Value Proposition & Solutions)

The OmniPact Protocol is not merely a minor adjustment to existing transaction models, but a complete overhaul of commercial trust mechanisms. We have distilled its core values into five pillars, using cutting-edge technology to precisely address the pain points of traditional transactions.

1.4.1 The Five Pillars of Value

1. Trustless Security

"Don't trust verify."

- Non-Custodial Mechanism: Unlike traditional platforms, OmniPact protocol (and its developers) have no authority to access or divert user funds. The funds are locked in an audited open-source smart contract, and can only be transferred when specific cryptographic conditions are met, such as mutual multi-signature confirmation or arbitration results being recorded on the blockchain.
- Atomic Settlement: This mechanism links payment and delivery within a single blockchain transaction, enabling digital asset trading to achieve the 'one hand pays, one hand delivers' principle. Mathematically, it eliminates default risks.

2. Decentralized Justice

"Replace authoritarian adjudication with community consensus."

- The Decentralized Arbitration Network (DAN) is introduced. When disputes arise, the decision-making authority is no longer vested in a black-box service provider, but rather in randomly selected, impartial global jurors who vote according to game theory (Shelling Point Mechanism). This ensures the fairness, transparency, and non-bribery nature of the rulings.

3. Universal Compatibility

One agreement, all things are traded.

- OmniPact transcends the limitations of DeFi confined to token swaps. Through standardized interfaces and modular design, we support:
    - Standard assets: ERC-20, NFT, etc.
    - Non-standard assets: social media accounts, game equipment, and domain names.
    - Real World Assets (RWA): Supported by oracles and legal agreements on the blockchain, it facilitates complex real-world transactions including service outsourcing, physical goods, and intellectual property.

4. Extreme Cost Efficiency

"Say goodbye to 20% intermediary exploitation."

- By automating operations to replace manual processes, OmniPact minimizes transaction friction. Its base protocol charges only 0.1% -1% (and is even free under certain conditions), significantly lower than traditional platforms' 10%-300% fees. This makes micro-profit industries, high-frequency trading, and cross-border small payments economically viable.

5. User Sovereignty and Data Portability

"Your reputation is yours alone."

- The on-chain reputation system built on SBT (Soulbound Token) complies with EIP-5192, converting users 'transactional behavior into tamper-proof credit credentials. These credentials are user-owned and cross-platform. A strong fulfillment record serves as a' golden credential' in the Web3 ecosystem, enabling lower collateral ratios and higher commercial premiums.

1.4.2 Core Pain Points vs. OmniPact Solutions

To demonstrate how OmniPact tackles real-world challenges, we systematically compare the pain points of traditional centralized platforms (Web2) and raw crypto transactions with our solution:

Core Pain Points	Scenario example	OmniPact Solution	Results (Outcome)
funds security risk	The platform absconded with the funds; the buyer maliciously withdrew the payment after receiving the goods; the seller blocked the buyer and fled after receiving the payment.	Smart Contract Escrow: Funds are locked on the blockchain and can only be released through multi-signature verification or arbitration.	Zero risk of fund diversion: The capital cannot be unilaterally misappropriated under the established financial framework.
high commission	Upwork charges a 20% service fee; cross-border remittance incurs a 5% fee.	The automated protocol layer eliminates intermediary labor costs, charging only minimal Gas fees and protocol fees.	<1% Fee: Profits are shared between the two parties.
Lack of a foundation of trust	Anonymous OTC transactions; too afraid to make the first payment or distribute the coins.	On-chain Reputation System (Reputation SBT): A protocol-embedded credit scoring mechanism that visualizes historical fulfillment data.	Verifiable trust: Even strangers can quickly establish trust based on data.
Dispute resolution black box	The centralized platform shows no reason to favor buyers; customer service responds slowly and makes arbitrary decisions.	Decentralized Arbitration Network (DAN): A global distributed jury system with on-chain evidence storage and transparent processes.	Procedural justice: Fair adjudication based on evidence and consensus.
Physical items or services cannot be added to the blockchain	I booked the design service, but the design he provided was completely black. How could the smart contract even recognize it?	Hybrid Verification: A system combining oracle data input with DAO's manual arbitration as a fallback mechanism.	Full-scenario coverage: Supports both subjective services and physical transactions.

1.5 Target Audience & Application Boundaries

The OmniPact Protocol is designed to serve as a universal adapter for global value exchange. Our design philosophy of 'bottom-layer neutrality and top-layer diversity' ensures our user base spans a broad spectrum, from individual retail users to enterprise-level institutions.

1.5.1 Multi-dimensional Target User Segments

We divide ecological participants into three core tiers:

1. C-end: Independent Traders & Creators (Individual Traders & Creators) These are the cornerstone users of the protocol, prioritizing low entry barriers and security.

- Web3 natives / Degen: Users who frequently engage in OTC token exchanges, private NFT transactions, and whitelisted account trading. They urgently seek to eliminate reliance on 'intermediary guarantees'.
- Digital nomads and freelancers: Global developers, designers, and translators. They face challenges in cross-border payments and the risk of client refusal (Rug Pull).
- Cross-border e-commerce buyers/sellers: Individuals involved in overseas shopping, purchasing on behalf of others, or second-hand luxury goods transactions need to address trust-building and payment processing challenges.

2. B-side: DAO & SMB. This segment drives high-frequency protocol liquidity, prioritizing automation and regulatory compliance.

- DAO (Decentralized Autonomous Organization): It requires issuing rewards (Bounty) to contributors, conducting asset swaps (Token Swap) with other DAO, and utilizing multi-signature and conditional execution tools.
- OTC Desks: Major crypto asset traders that utilize OmniPact as a custodial layer to mitigate security risks of their holdings.
- Supply chain enterprises: Implement smart contracts to automate cash-on-delivery settlements and optimize cash flow.

3. D-side: Developers & Partners This segment serves as the moat builder of the protocol, prioritizing user-friendliness and stability.

- Web3 marketplace/platforms: such as decentralized platforms like Fiverr and eBay's competitors. They don't need to develop complex trust guarantee systems themselves; simply integrating the OmniPact SDK grants them financial-grade guarantee capabilities.
- Wallet service providers: Embed a 'transfer guarantee' feature into wallets as a value-added service for users.

1.5.2 Application Scenario Spectrum

OmniPact application scenarios follow a spectrum ranging from simple to complex based on delivery verification difficulty. The protocol provides tailored solutions with varying depth of coverage for different regions.

A. Native On-chain Assets — [fully automated]

- Scenarios: Token swaps, NFT trading, and DeFi position transfers.
- Mechanism: Atomic Swap. The smart contract directly reads on-chain data with 100% accuracy, eliminating manual arbitration and enabling instant settlement.

B. Off-chain Digital Delivery — [Evidence Storage + Arbitration]

- Scenarios: code delivery, UI design drafts, API keys, game account passwords, and private community access tickets.
- Mechanism: The seller uploads the encrypted deliverable or hash value to the chain/IPFS. If the buyer files a fraud complaint, the arbitrator decrypts and verifies the deliverable's

authenticity.

C. Real World Assets & Services — [oracle + arbitration]

- Services include: physical e-commerce (express delivery), RWA asset tokenization (real estate/vehicle leasing), consulting services, and offline errand delivery.
- Mechanism: Access logistics API (e.g., Chainlink) to track physical conditions; integrate data from IoT devices; and ultimately rely on decentralized arbitration panels to resolve subjective disputes (e.g., 'poor service attitude' or 'product wear').

1.5.3 Scope Boundaries & Negative List

To maintain the protocol's efficiency and focus, it is essential to clearly define scenarios where OmniPact is not applicable or not yet supported. This reflects our honest recognition of technical limitations.

Applicable dimension	**The Sweet Spot**	**Out of Scope**	**Rationale**
trade frequency	**low and medium frequency trading** (Low/Medium Frequency)	**high frequency quantitative trading** (HFT / Market Making)	Block Time and Gas fees are not suitable for millisecond-level matching.
transaction amount	**Medium to high value deals ($50-$10M+)** Commercial contracts, bulk commodities, and wage settlements.	**Micro payments (<$1)** Reward ads, pay-per-click.	The gas fee and arbitration costs (Juror Fee) may exceed the transaction's value.
compliance	**Compliant and legal goods/services** assets within the scope of free trade	**Prohibited Goods/Washing Money** Dark web transactions and sanctioned activities.	While the protocol layer remains unlicensed, the front-end application layer will enforce rigorous risk control and a strict blacklist mechanism.
instantaneity	**delayed delivery / secured transaction** (Escrow / Milestone)	**Instant transfer** (Instant Transfer)	Our core value is "locking and unlocking" rather than a simple payment channel.

OmniPact is neither a payment solution for daily coffee purchases nor a high-frequency matching engine to replace Binance. It is a settlement infrastructure specifically designed for non-immediate trust scenarios. When parties lack mutual trust and transactions involve time gaps or delivery uncertainties, OmniPact serves as the only viable solution.

1.6 White Paper Reading Guide (Guide to Reading This Whitepaper)

The OmniPact Protocol is a sophisticated system that integrates cryptography, game theory, financial engineering, and business logic. To help readers from diverse backgrounds efficiently grasp the core concepts, we've curated the following reading roadmap tailored to your role and interests:

1.6.1 Role-Based Navigation

Whether you're a return-seeking investor or an integration-focused developer, the table below provides the best reading guide:

Your Role	Key Focus	Recommended Chapters
Investors / Analysts (Investors & Analysts)	Business model, token value capture, growth potential	Chapter 1 (Core Overview) Chapter 6 (Token Economics-$PACT) Chapter 8 (Roadmap and Milestones)
Developer / Geek (Developers & Geeks)	Technical architecture, contract logic, and security mechanisms	Chapter 2 (Technical Architecture) Chapter 3 (Arbitration Consensus Mechanism) Chapter 9 (Security and Audit)
Partners / Businesses (Partners & Merchants)	Implementation scenarios, business processes, and access methods	Chapter 4 (Business Mechanisms and Workflows) Chapter 5 (Application Scenario Overview) Chapter 7 (Governance and

Your Role	Key Focus	Recommended Chapters
		DAO)
Legal / Compliance Expert (Legal & Compliance)	jurisdiction, dispute resolution, regulatory adaptability	Chapter 3 (Decentralized Arbitration) Chapter 10 (Legal Disclaimer)

1.6.2 Key Terminology

Before diving into the details, understanding the specific terminology in the OmniPact ecosystem will help you better grasp the protocol's logic.

- Smart Escrow (intelligent escrow): The core mechanism of OmniPact, which locks assets and automatically releases funds based on predefined conditions.
- DAN (Decentralized Arbitration Network): A decentralized arbitration network functioning as a 'court on the blockchain,' composed of globally distributed nodes.
- Juror (Arbitrator/Jury Member): Nodes that stake $PACT tokens and participate in dispute resolution earn rewards by contributing their expertise.
- SBT (Soulbound Token): A non-transferable and non-tradable token that stores a user's reputation history, compliant with ERC-5114 protocol.
- Atomic Settlement (atomic settlement): A payment and delivery process completed in a single transaction, where the entire transaction is either fully successful or fully unsuccessful, with no intermediate states.
- RWA (Real World Assets): Real-world assets, which are digital tokens representing physical-world properties, goods, or rights on the blockchain.

2. Technical Architecture

2.1 Overall Architecture Design

2.1.1 Protocol Layer: The Anchor of Trust

The protocol layer serves as the core of OmniPact, built directly on the underlying blockchain (Layer 1/Layer 2). It consists of a set of tamper-proof smart contracts that handle all logic related to fund security and core state changes. This layer adheres to the 'principle of minimization,' retaining only the most essential settlement functions to ensure security.

- Universal Escrow Contract: This serves as the protocol's' treasury'. Built on our OES (Omni Escrow Standard) framework, it governs all state transitions throughout the transaction lifecycle—creation, locking, release, and refund. The non-custodial design ensures that no administrator (including OmniPact team) can move funds without

multisignature verification or arbitration credentials.

- The Arbitration Consensus Interface (ACI) serves as the logical gateway to decentralized arbitration networks (DAN). It verifies the weight of arbitrators' signatures, calculates voting results, and converts the final ruling into on-chain execution instructions.
- Settlement Engine: Manages complex fund allocation logic. Enables atomic distribution of a single transaction to multiple recipients (e.g., sellers, referents, agreement treasuries, arbitrators).

2.1.2 Service Layer: Connectivity & Extension

The service layer serves as the protocol's' brain and limbs', acting as middleware between on-chain code and the off-chain world. It addresses the limitations of smart contracts, such as their inability to store large data or access external data.

- The Omni-Link Oracle Gateway acts as an adapter, converting real-world trigger events (e.g., DHL logistics confirmation messages, flight delay data, API interface statuses) into Boolean values (True/False) readable by smart contracts, enabling automated fulfillment of RWA assets.
- Decentralized Evidence Storage: Utilizes IPFS and Arweave to store rich media evidence during transactions, including chat screenshots, scanned contracts, and design source files. The service layer only records the file's Content Hash on the blockchain, ensuring tamper-proof and permanently traceable evidence.
- Reputation Identity System (RIS): Analyzes users 'on-chain behavior to generate Omni-ID and its corresponding SBT (Soul Bond Token). It calculates users' Credit Score in real time, providing risk assessment data for upper-layer applications.

2.1.3 Application Layer: Ecosystem Interfaces

The application layer serves as OmniPact face'—the interface through which users and developers interact directly. It transforms complex underlying logic into user-friendly tools.

- OmniPact SDK & API: A standardized software development kit (SDK) for third-party developers. Whether you're building a Web3 NFT marketplace or a Web2 e-commerce platform, OmniPact collateral services can be activated with just a few lines of code—no need to develop a contract from scratch.
- Official dApp interface: A no-code interface for end users. Through the graphical interface, users can create collateral contracts, upload evidence, and initiate arbitration with one click.
- Vertical Solutions: Customized templates for specific scenarios (e.g., freelancing, OTC, supply chain). For example, the "OTC Template" automatically configures a 24-hour cooling-off period, while the "Outsourcing Template" sets up milestone-based phased payments.

2.2 Core Components of the Protocol Layer

2.2.1 OES Standardized Escrow System (The Omni Escrow Standard System)

The OmniPact protocol layer is not merely a collection of simple scripts, but a comprehensive standard based on Finite State Machine (FSM) theory, which we call the Omni Escrow Standard (OES).

OES is designed to serve as the 'transaction container' for the Web3.0 era. Just as ERC-20 established token standards and ERC-721 defined NFT standards, OES sets the benchmark for 'conditional value exchange'.

1. Finite State Machine Model

The OES contract mitigates transactional uncertainty through rigorous state transitions. Each security contract instance (Instance) operates within an irreversible lifecycle, where state changes are exclusively triggered by predefined cryptographic signatures or oracle signals.

- Initialization (Initialized): The buyer and seller use a hash algorithm to write the transaction terms (price, delivery description, deadline) onto the blockchain, generating a unique Agreement Hash.
- Locked: Buyer funds (Token/NFT) are injected into the contract. At this point, the funds enter a frozen state, belonging neither to the buyer nor the seller, but to the protocol logic.
- Active/Delivery: The seller submits Proof of Delivery, such as the hash of the logistics tracking number, Commit hash from the codebase, or oracle signals from off-chain services.
- Dispute/Optional: If the buyer refuses to confirm or file a claim, the contract will be automatically suspended, transferring control to the arbitration module (DAN) for external adjudication.
- Finalized:
    - Path A (Happy Path): The buyer confirms or the system automatically confirms upon timeout → Funds are released to the seller.
    - Path B (Ruling Path): The settlement is executed based on the arbitration outcome (e.g., 80% refund and 20% compensation).

2. Non-Custodial & Atomic Security

OES has completely abandoned the 'fund pool model' of Web2.0 platforms.

- Isolated Vaults: Each transaction deploys or generates a unique virtual vault address via a Factory Contract. This physical isolation between transactions eliminates risks of platform fund misappropriation or bank runs.
- Atomic Settlement: OES ensures that 'payment' and 'delivery verification' are completed within the same blockchain transaction. If the conditions are not met, funds remain frozen; once satisfied, the funds are irrevocably transferred.

3. Metadata Extensibility

To support the "All Things Guarantee" initiative and build the world's first Web3.0 universal trust infrastructure, OES has designed a flexible metadata interface called the _data field.

- Structured data injection: Supports not only simple transfer guarantees but also complex business logic parameters.
    - Example 1 (Physical E-commerce): Inject {logistics-provider: "DHL", tracking_no: "123456"} to automatically trigger contract-based logistics oracle polling.
    - Example 2 (Development Outsourcing): Inject {milestone_1:

"30%",github_repo: "xyz"} with phased funding release.

- Privacy encryption: For sensitive commercial contract details, OES supports Hash-only (only hash value) on-chain storage. The original data is stored on IPFS using asymmetric encryption, with decryption private keys held exclusively by the transaction parties and arbitrators.

4. Timelock and Circuit Breaker Mechanism

- Auto-Release Timelock: To prevent buyers from maliciously refusing to confirm and causing funds to be locked, each contract must include an expiration_time. If the time expires without any dispute, the contract defaults to buyer satisfaction, and the funds are automatically released.
- Emergency Pause: When detecting a reorganization attack on the underlying public chain or abnormal oracle quotes, the DAO can trigger circuit breakers to suspend all contract liquidity outflows, ensuring user asset security.

2.2.2 SBT Reputation Identity System

While blockchain's anonymity safeguards privacy, it simultaneously creates significant trust friction. To strike a balance between anonymity and trust, the OmniPact protocol layer incorporates a decentralized identity system called Omni-ID, with its underlying medium being the non-transferable Omni Soulbound Token (O-SBT).

This is not merely a points system, but a 'credit passport' in the Web3.0 economy.

1. O-SBT: Non-transferable Credit Assets

O-SBT complies with the EIP-5192 (Minimal Soulbound NFT) standard. It is a non-fungible token (NFT) permanently linked to a user's wallet address, which is non-transferable and non-tradable.

- Uniqueness and Sybil Resistance: Each address can only mint one O-SBT. This mechanism tracks users' full lifecycle activities across the OmniPact network. Given that reputation accumulation requires substantial time and real-money transactions, it significantly raises the cost for malicious actors to commit fraud by creating numerous fake accounts.
- Multidimensional Metadata: O-SBT not only tracks a single score, but also captures multi-dimensional performance capability tags through metadata fields. For example:
    - Role: [Buyer, Seller, Arbitrator]
    - Domain: [Design, Logistics, OTC] (Areas of expertise)
    - Success_Rate: 98.5%
    - Total_Volume: $1,250,000

2. Dynamic Reputation Scoring Algorithm

The protocol introduces a set of on-chain quantification models (Reputation Engine) to dynamically update the reputation value of O-SBT based on users' actual performance in OES guarantee contracts. Scoring dimensions include but are not limited to:

- Performance quality: Number of successful and undisputed transactions.
- Dispute log: The frequency of losing cases as a defendant (which would cause a cliff-like drop in credibility score).
- Funding scale: Total amount of guarantees processed (lower weighting to prevent monopolization by the wealthy).

- Time weighting: Recent transactions carry higher weight than future ones, ensuring the timeliness of evidence.
- Social verification: peer reviews or endorsements from other highly trusted users (Web of Trust).

3. Zero-Knowledge Proof Privacy Preservation (ZK-Privacy Preservation)

We recognize that users' transaction history (e.g., purchases and sales amounts) constitutes highly sensitive commercial privacy. To address this, the Omni-ID system incorporates zk-SNARK (zero-knowledge succinct non-interactive knowledge proofs) technology.

- Non-disclosure verification: When users need to verify their credibility on third-party platforms (e.g., lending agreements or job platforms), they are not required to disclose specific O-SBT raw data.
- Application scenario: Users can generate a zero-knowledge proof (ZKP) to demonstrate that their OmniPact credit score exceeds 700 and they have no fraud history, without disclosing specific account balances or transaction history.

4. Credit Utility: The Monetization of Trust

In the OmniPact ecosystem, high credibility is not just an honor—it's a tangible economic privilege (Economic Privilege).

- Service fee waiver: Premium users (Level 5+) qualify for 50%-100% discount on agreement service fees.
- Airdrop rewards: High-scoring users may receive PACT airdrops at random intervals.
- Under-collateralized Escrow: For merchants with top-tier credit ratings, the agreement permits lower margin requirements for secured transactions, even enabling a 'trade-first, settle-later' credit payment model.
- Arbitrator Qualification: Only users with O-SBT credit scores meeting a specific threshold are eligible to serve as jurors on the DAN (Decentralized Arbitration Network) and earn arbitration fees.

2.2.3 Decentralized Arbitration Module (DAN)

The DAN (Decentralized Arbitration Network) serves as the 'final arbiter' in the OmniPact protocol and is the core component for resolving Subjective Disputes. When smart contracts cannot determine outcomes through simple 'true/false' logic (e.g., a buyer claims a design draft is 'lacking creativity' or a physical item has 'scratches'), the DAN is activated.

DAN is not dependent on a small group of elite judges, but rather operates on the principles of 'collective intelligence' and 'cryptoeconomic game theory'.

1. Arbitrator Selection Mechanism: VRF-Based Random Sampling

To prevent conflicts of interest and witch attacks, the selection of jurors must be entirely random and unpredictable.

- Staking: Users must stake a specified amount of $PACT tokens in the protocol to activate their arbitrator status. The staking amount reflects their 'weight of rights and responsibilities'.
- Chainlink VRF (Verifiable Random Function): When disputes arise, the protocol invokes Chainlink VRF to generate truly random numbers and selects 3-7 arbitrators from a qualified pool for preliminary arbitration.

- Probability weight: The probability of being selected is proportional to the number of staked $PACT. This ensures that those with greater "system benefits" are more motivated to uphold the system's fairness.

2. The Core of Game Theory: The Schelling Point Consensus

How can we ensure that anonymous arbitrators, who are not in communication with each other, can make honest judgments? DAN employs a game mechanism based on the Schelling Point.

- The core logic: An arbitrator's objective is to predict the voting tendencies of fellow arbitrators. In the absence of communication, the natural focal point (Schelling Point) becomes the truth of facts, since truth is singular while falsehoods are manifold.
- Incentive & Slashing Mechanism:
    - Coherence Reward: Arbitrators whose votes align with the majority will share the arbitration fees paid by both disputing parties and receive an additional $PACT reward as per the agreement.
    - Deviation Slashing: Arbitrators who cast dissenting votes (i.e., minority votes) are deemed to have made a 'judgment error' or committed misconduct. The portion of $PACT they staked will be deducted (Slash) and redistributed to the winning arbitrator.
- Results: In order to keep the principal and earn the profit, the rational arbitrator must examine the evidence carefully and make the most fair judgment in accordance with the public cognition.

3. Anti-cheating Voting Process: Commit-Reveal Scheme

To prevent arbitrators from copying each other's voting results (commonly known as' vote-picking'), DAN implemented a two-stage voting mechanism:

- Phase One: Commit Phase: The arbitrator locally generates a voting result (either in favor of the buyer or seller) and a random salt value, then submits their respective hash values to the blockchain. At this stage, external observers can only see garbled data and cannot discern the voting intent.
- Phase Two: Reveal Phase: The system initiates the reveal phase after all arbitrators submit their hashes. Each arbitrator then provides the original voting content and corresponding salt value. The smart contract verifies the hash matches and tallies the final votes.

4. Appeal and Escalation System

To address occasional erroneous rulings (such as those by first-instance arbitrators due to insufficient expertise), DAN has implemented a multi-stage appeal mechanism.

- Doubling Stake: If dissatisfied, the losing party may appeal but must pay a higher arbitration fee than the previous round.
- Enlarged jury: The number of arbitrators drawn for each round of appeals doubles (e.g., 3 → 7 → 15 →...). As the number increases, the probability of bribery or statistical bias decreases exponentially.
- Final Review Panel: If the above methods fail to rectify the erroneous judgment, we will initiate the Final Review Panel intervention. This panel is composed of users with high O-SBT holdings and high staking activity.
- Finality: When the appeal cost exceeds the disputed amount, the system will not accept

further appeals. The current result becomes the final ruling (Final Ruling), and the smart contract will automatically execute the profit distribution.

2.2.4 Cross-Chain Compatibility Module

OmniPact's vision is to connect global value, requiring the protocol to be' chain-agnostic.' Our cross-chain compatibility module eliminates liquidity fragmentation across different blockchains (Layer 1 and Layer 2), enabling users to initiate guarantees on any supported chain and complete settlements on another chain.

1. Architecture based on Interoperability Standards

For security and decentralization, OmniPact does not build a proprietary closed cross-chain bridge. Instead, it functions as an 'application-layer protocol' that integrates industry-proven universal messaging standards (e.g., Chainlink CCIP, LayerZero, Wormhole).

- Security Reuse: By leveraging the decentralized oracle networks (DONs) and relays of these infrastructures, OmniPact inherits their multi-billion-dollar security consensus, effectively avoiding the private key leakage risks inherent in self-built single-point bridges.

2. Cross-Chain State Synchronization

OmniPact innovatively decouples 'guarantee logic' from 'asset settlement', enabling atomic state synchronization across heterogeneous blockchains.

- Source Chain operation: The buyer locks assets in an OmniPact contract on Chain A (e.g., Ethereum). The contract generates a cross-chain message packet containing transaction metadata, which is then sent to the target chain via a relayer.
- Destination Chain Response: Upon receiving and validating the message, the OmniPact contract on Chain B (e.g., Solana) generates a local mirror guarantee. The seller initiates delivery when the status changes to 'locked' on Chain B.
- Atomic termination: When delivery on Chain B is confirmed (or arbitration completed), the Chain B contract sends a "release instruction" to Chain A. The funds on Chain A are then unlocked for the seller (or exchanged for Chain B assets via a cross-chain liquidity pool).

3. Unified Liquidity Routing

The module features an Aggregated Liquidity Engine, which resolves asset fragmentation in cross-chain payments.

- Pay in Any Token: Buyers can use USDC on Arbitrum to make payments, while sellers receive USDT on BNB Chain.
- Automated Exchange: The cross-chain module automatically invokes underlying DEX aggregators (e.g., 1inch, Uniswap X) to execute optimal-rate conversions during fund transfers, while Gas Abstraction simplifies the process. Users can perform these operations without holding native tokens of the target chain.

4. Safety Isolation and Circuit Breakers

To address potential cross-chain systemic risks, we have developed the Bulkhead Pattern.

- Rate Limiting: A system that imposes time-bound caps on cross-chain fund transfers for each chain and asset type.
- Independent Fault Tolerance: When a public chain (e.g., a Layer 2 network) experiences a failure or rollback, the cross-chain module automatically disconnects from it, ensuring

OmniPact protocols on other chains remain unaffected and funds stay secure.

2.3.1 Decentralized Evidence Storage Solution: Omni-Vault (Decentralized Storage)

In commercial guarantee transactions, the volume of data generated often far exceeds the capacity of blockchain ledgers. High-definition design drafts, scanned copies of complex legal contracts, and months-long chat logs and logistics records would incur astronomical Gas costs if directly added to the blockchain.

To this end, OmniPact developed Omni-Vault, a hybrid decentralized storage layer based on Content-Based Addressing. It achieves an efficient balance between 'light on-chain settlement and heavy off-chain storage'.

1. Off-chain Storage, On-chain Verification

Rather than writing raw files to smart contracts, Omni-Vault employs a Hash Anchoring mechanism.

- Data Fingerprint (CID): When a user uploads a file as evidence, the system calculates its SHA-256 hash (CID) locally.
- Status lock: Only this 32-byte hash value will be written to the _evidenceHash field of the OES guarantee contract.
- Mathematical Tamper-Proofing: The hash function's avalanche effect ensures that any alteration to a single pixel or punctuation mark in the original file will cause the hash value to change drastically, making it impossible for smart contracts to validate. This guarantees the authenticity and non-repudiation of evidence during arbitration.

2. Dual-Track Storage Strategy

To balance access speed and storage persistence, Omni-Vault integrates two underlying protocols: IPFS (InterPlanetary File System) and Arweave (Permanent Storage Network).

- Hot Tier (IPFS):
    - Scenarios: Temporary files, instant chat records, and drafts during transaction negotiations.
    - Features: Fast access speed and extremely low cost, ideal for frequently read and updated data.
- Cold Tier (Arweave):
    - Applicable scenarios: signed final contracts, arbitral tribunal's final ruling (Final Ruling), and critical acceptance certificates.
    - Key feature: "One-time payment, permanent storage". The OmniPact protocol automatically pays storage fees to Arweave nodes via smart contracts, ensuring these critical legal documents remain accessible for over 200 years. Even after the OmniPact team's dissolution, the data will continue to exist.

3. Commercial-grade privacy encryption (Client-side encryption)

Given the sensitivity of commercial secrets (e.g., source code, customer lists, and proprietary formulas), Omni-Vault enforces a strict 'encrypt before upload' policy.

- Asymmetric encryption tunnel: All files are automatically encrypted with the counterparty's public key and the arbitration tribunal's public key before leaving the user's device.
- Permission isolation:
    - On public IPFS networks, these data appear as encrypted blobs (Encrypted

Blobs).
    - Only buyers, sellers, and selected arbitrators with the corresponding private key can decrypt and view the plaintext.
    - Even storage nodes or OmniPact developers cannot access the file contents.

4. Data Availability Assurance

To prevent data withholding by storage nodes, Omni-Vault implements a challenge mechanism:

- Periodic Proof of Retention (PoR): The protocol randomly initiates data integrity challenges (PoR) to storage nodes.
- Storage incentives: A portion of users' storage fees goes into the incentive pool for storage nodes, with $PACT rewards awarded only to nodes that consistently prove their data is online.

2.3.2 Omni-Link Oracle Network: A Full-Scale Predictive Analytics Platform

Smart contracts are inherently closed sandbox systems that cannot actively perceive the state of the off-chain world. To achieve universal collateralization, OmniPact developed Omni-Link—an decentralized all-dimensional data gateway. It acts as a translator between complex real-world events and blockchain's deterministic logic.

1. Generalized Data Adaptation

Unlike traditional DeFi oracles that focus solely on financial price data, Omni-Link specializes in capturing and validating business transaction data. It supports standardized HTTP/HTTPS request encapsulation, enabling seamless integration with 99% of global RESTful API.

- Logistics tracking: Aggregates data from DHL, FedEx, UPS, and national postal services, delivering real-time parcel status updates (collected, in transit, or signed for) to the blockchain.
- Digital service metrics: Integrating GitHub API (code commit status), Uptime Robot (server uptime), and Figma API (design update frequency) to provide objective evaluation criteria for SLA (Service Level Agreement) guarantees.
- Digital identity tracking: Monitoring Twitter/X retweets and Discord identity changes to automatically distribute marketing rewards.

2. Multi-Source Aggregation and Decentralized Verification

To prevent single data sources from being tampered with or experiencing failures (Single Point of Failure), Omni-Link employs a DON (Decentralized Oracle Networks) architecture.

- Multi-node consensus: A data request (e.g., "Is delivery tracking number 123 signed for?") is distributed to multiple independent oracle nodes.
- Aggregation Logic: Nodes collect data from multiple sources (e.g., logistics websites, third-party aggregation platforms, or local scanners). The smart contract only accepts the data and triggers fund release when over 2/3 of nodes return a Consensus Answer.
- Exception removal: Data deviating from consensus is marked as "dirty data", and the node submitting this data will be penalized with a credit downgrade.

3. Off-Chain Computation and Privacy Protection

For complex validation logic (e.g., 'verify if this Python code can pass 50 test cases'), running

it directly on-chain is prohibitively expensive and constrained by Gas Limit.

- Omni-Link Compute: Enables developers to create custom verification scripts (Adapter Scripts) for execution in the Trusted Execution Environment (TEE).
- On-chain result: Upon script execution, the smart contract only records the minimal Boolean result (Pass/Fail) and execution proof. This approach safeguards business logic privacy while drastically reducing Gas costs.

4. Bi-directional Event Stream

Omni-Link not only reads data but also writes operations.

- Callback Trigger: When a collateralized contract on-chain undergoes a status change (e.g., arbitration concluded or buyer wins), Omni-Link can execute off-chain Web2 actions, such as automatically sending a refund notification email or invoking API to revoke the buyer's SaaS software license.

2.3.3 Device Binding & Security Authentication: IoT-Anchor

The biggest challenge in RWA tokenization lies in the oracle fracture: how to ensure on-chain assets (NFT) uniquely correspond to and control real physical devices? OmniPact developed the IoT-Anchor protocol, which assigns non-forgery on-chain identities to physical devices through a combination of software and hardware solutions.

1. PUF-Based Hardware Fingerprint

Traditional device identifiers (e.g., MAC addresses) are vulnerable to software tampering or cloning. IoT-Anchor implements PUF (Physical Unclonable Function) technology to address this vulnerability.

- Microscopic physical variations: Leveraging atomic-level random physical variations generated during chip manufacturing, it creates a unique, non-replicable 'biological fingerprint'.
- Endogenous Private Key (Key Generation): The device's private key is generated directly within the chip by the PUF physical properties and cannot be exported. Even if a hacker disassembles the device, they cannot obtain the private key. This ensures that every device sending data to the OmniPact contract is a uniquely authenticated original from the manufacturer.

2. DID IoT Identity Standard (DID for Things)

OmniPact assigns a W3C-compliant DID (Decentralized Identifier) to each certified IoT device.

- Device as a Wallet: Each smart device has its own independent contract account on the blockchain.
- Self-executing transactions: Devices can directly execute transactions. For example, when a smart lock receives a token from a tenant, it automatically signs a 'lock opened successfully' message on the blockchain as proof of fulfillment, without human intervention.

3. Sign-to-Live Heartbeat

To monitor the status of leased assets (preventing equipment theft or offline status), IoT-Anchor has implemented a robust heartbeat verification mechanism.

- Challenge-Response: The OmniPact protocol periodically sends random numbers (Nonces) to the device.

- Proof of existence: The device must sign the random number with the PUF private key and return it. If the smart contract does not receive a valid signature within the specified time window, the device will be automatically marked as "abnormal/offline" and trigger the circuit breaker mechanism (e.g., suspending profit distribution to the lessor or initiating insurance claims).

4. Physical Tamper-Proofing and Data Integrity

To counteract malicious physical attacks (e.g., sensor tampering), IoT-Anchor employs multidimensional sensor data fusion verification.

- Environmental Awareness: The device is equipped with an accelerometer, GPS, and a light sensor. If it detects abnormal vibrations or a case opening (triggering a light sensor change), it immediately signs and broadcasts an 'intrusion' alert to the blockchain. This record permanently impacts the asset's valuation and reputation.
- Data authentication: All operational data (e.g., mileage, electricity generation) uploaded by devices are equipped with hardware signatures, ensuring the data source remains tamper-proof and effectively addressing the persistent issue of data falsification in supply chain finance.

2.3.4 Data Privacy Protection Scheme: Privacy Matrix

In the business world, privacy is not merely a right but a core asset. OmniPact recognizes that a protocol failing to protect trade secrets cannot gain mainstream adoption. To address this, we developed the Privacy Matrix—a tiered privacy protection framework that integrates cryptography with hardware isolation.

1. Client-Side Asymmetric Encryption

OmniPact adheres to the principle of 'on-chain encryption.' All sensitive data (such as contract details, chat records, and delivery source files) must be encrypted before leaving the user's device.

- Multi-party key distribution: The system automatically aggregates the buyer's public key, the seller's public key, and (potentially) the arbitration network's public key to perform hybrid encryption on the data.
- Permission isolation:
    - From a public perspective: In IPFS/Arweave networks and blockchain systems, data is represented as a string of meaningless encrypted blobs (Encrypted Blobs).
    - Authorization view: Only the parties with the corresponding private key can decrypt and view. If the case enters arbitration, the selected arbitrator will receive temporary decryption authorization, which automatically expires after the case concludes.

2. Zero-Knowledge Proof Verification (ZK-Proofs Verification)

For scenarios requiring public verification without exposing plaintext, OmniPact incorporates zk-SNARK technology to enable zero-leakage verification.

- Asset solvency verification: The buyer can generate a ZK proof to demonstrate to the seller that 'my wallet balance exceeds the transaction amount,' without disclosing the total account assets to the entire network.
- Creditworthiness verification: Enterprise users can verify that their Omni-ID credit score

exceeds 800 and they are not on the AML blacklist, without disclosing specific transaction history or amounts.

3. Trusted Execution Environment (TEE) Privacy Computing

When the oracle requires access to sensitive external data—such as querying inventory in an enterprise's ERP system or inspecting code in a private GitHub repository—we utilize Trusted Execution Environment (TEE) technology like Intel SGX.

- Hardware-level black box: The data validation script operates within the CPU's encrypted enclave. Even the physical server owner (node operator) cannot access the memory data.
- Result minimization: TEE only outputs computation results (e.g., True/False) to the blockchain, never disclosing raw data (e.g., passwords or source code).

4. Compliance and "Crypto-Shredding" (GDPR Compliance Measures)

Blockchain's immutable nature inherently conflicts with the' Right to be Forgotten 'under the EU's General Data Protection Regulation (GDPR). OmniPact resolves this compliance challenge through its 'cryptographic erasure' mechanism.

- Logical deletion: When a user requests to delete data, the hash value on the chain and the encrypted data block in the storage network cannot be physically deleted, but the user can destroy (or request protocol destruction) the private key used to decrypt the data.
- Mathematical erasure: When the private key is lost, the remaining encrypted data becomes mathematically uncrackable, rendering it meaningless random noise. This fulfills the 'data unrecoverable' compliance requirement from both legal and technical perspectives.

2.4.1 Front-end dApp/App Module: No-Code Escrow Center

The OmniPact official decentralized application (dApp) and mobile app serve as the protocol's primary gateway for end users. Featuring a wizard-style interaction design, it enables users to create legally binding and technically enforceable on-chain guarantee contracts within one minute without coding.

1. Smart Transaction Wizard

We have eliminated complex parameter entry and instead provide a scenario-based creation process:

- Scenario Templates: The system includes over 20 industry-standard templates that have been audited.
    - OTC template: Predefined ERC-20 token swap logic with automatic exchange rate matching.
    - Service outsourcing template: Features a preset 'Milestone' function, supporting phased fund release (3-3-4).
    - E-commerce template for physical goods: Requires mandatory logistics tracking number entry and automatically integrates with Omni-Link logistics forecasting system.
- One-Click Deploy: Users simply enter key fields like "Counterparty Address", "Amount", and "Delivery Deadline". After confirming, the system automatically converts these business requirements into OES smart contract parameters and broadcasts them to the

blockchain.

2. Panoramic Asset Management Dashboard

Users can manage all active transactions through a single interface, similar to a Web3-style order center.

- Status visualization: Clearly display the current status of each transaction — pending, in fulfillment, pending confirmation, or in dispute.
- Key action trigger: Dynamically activate the action button based on the contract status. For example, the "Confirm Delivery and Release Loan" button in the buyer interface will only activate after the seller submits the deliverables to prevent accidental operations.

3. Dispute Resolution Center

When transactions encounter issues, this module provides a user experience similar to the after-sales backend of Web2 e-commerce platforms.

- Evidence Upload: Supports uploading images, PDF contracts, and chat records. The frontend automatically encrypts files locally (Client-Side Encryption) and uploads them to IPFS, only storing the hash values on the blockchain.
- Arbitration progress bar: Track case progress in real time (waiting for arbitrator selection-> evidence disclosure period-> voting in progress-> decision announcement). Users can clearly see when a result will be available.

4. Reputation Profile

This is the user's Web3 credit profile.

- SBT Visualization: Convert on-chain O-SBT data into intuitive radar charts, displaying user ratings across metrics like fulfillment rate, activity level, and asset size.
- Social Sharing: Create professional credit posters or custom links to help service providers showcase their reliability to clients when taking orders.

2.4.2 Open API/SDK Interface Specifications (Open API & SDK Standards)

The ultimate goal of OmniPact is to become the underlying "trust Lego" of the Web3 business ecosystem. To this end, we have released a Developer-First SDK and API suite, aiming to eliminate the complexity of interacting with smart contracts and empower third-party applications (such as e-commerce platforms, freelance markets, and OTC trading platforms) to quickly obtain financial-grade guarantee capabilities.

1. Full-stack OmniPact SDK (The OmniPact SDK Suite)

We provide SDK packages for mainstream programming languages (including TypeScript/JavaScript, Python, and Go), with built-in RPC communication, signature logic, and contract interaction details.

- Core feature encapsulation: Developers can handle complex collateralization processes by simply invoking standard functions, eliminating the need to write Solidity code.
    - Omni.createEscrow (params): Create a guarantee (automatically process parameter validation and hash to the chain).
    - Omni.lockFunds (escrowId): Trigger the wallet signature and lock the funds.
    - Omni.submitDelivery (proof): The seller submits the proof of performance (such as the logistics number or the file hash).
    - Omni.releaseFunds (escrowId): The buyer confirms receipt and triggers the

payment.
    - Omni.raiseDispute (reason): Upgrade to a dispute case and route it to the arbitration tribunal.
- Type Safety and Documentation: All SDK provide comprehensive TypeScript type definitions and detailed Swagger documentation, ensuring a bug-free and seamless development experience.

2. Read/Write API Gateway

To enhance data query efficiency, OmniPact provides a high-performance indexer API (Indexer API).

- GraphQL query interface: Enables developers to retrieve complex on-chain data with a single query using flexible syntax.
    - Example: Query all transactions with status 'Completed' for User A in the past 30 days, sorted by amount.
- Reputation API: Provides an open credit inquiry interface. Third-party DApp can directly call GET /user/:address/score to retrieve a user's Omni-ID credit score for risk control, access, or benefit distribution.

3. Event-Driven Webhooks System

To enable real-time synchronization of business status, OmniPact utilizes the Webhook push mechanism.

- Real-time callback: When critical events occur on the blockchain (e.g., successful fund locking or arbitrator's final ruling), the OmniPact node instantly sends an encrypted JSON payload to the developer-configured URL.
- application scenarios：
    - When the EVENT FUNDS LOCKED notification is received, the e-commerce platform automatically notifies the seller to ship the goods.
    - When the platform receives an EVENT_DISPUTE_RESOLVED notification, it automatically updates the order status to "Refund".

4. Security and Authentication

- API Key mechanism: Enterprise developers can apply for an API Key to obtain a higher call rate limit (Rate Limit).
- Signature Verification: All Webhook pushes include the X-Omni-Signature header. Developers can use the public key to verify that the message is authentic from OmniPact, preventing forged callback attacks.

2.4.3 Third-Party Integration Guide

The design philosophy of OmniPact is "Powered by OmniPact". We are not just a standalone application, but a foundational trust component that can be embedded in any business scenario. For different business models, we provide customized integration paths.

1. Web3 Native Platform Integration Solution (For Web3 Native Platforms)

- Target audience: NFT exchanges, decentralized over-the-counter (OTC) platforms, and GameFi accessory stores.
- Integration Mode: Embedded Escrow SDK.
- operation flow：

    1. Replace the payment button: The developer replaces the original "direct transfer" logic with a call to Omni.createEscrow .
    2. Transaction protection: When users click 'Buy NFT', funds are not directly transferred to the seller but locked in the OmniPact contract.
    3. Atomicized Delivery: The platform utilizes our callback interface to automatically trigger fund release upon detecting the NFT_transferRED event.
- Value: Instantly gain 'Anti-Rug Pull' capability, significantly boosting user retention and transaction volume on the platform.

2. Web2 Traditional E-commerce and Service Platforms (For Web2 eCommerce & Freelance)

- Target audience: Shopify standalone sites, WordPress plugin providers, and traditional freelance platforms (e.g., Upwork competitors).
- Integration mode: Payment plugins.
- operation flow：
    1. Plugin installation: Businesses can install the "OmniPact Secure Pay" plugin with one click in the Shopify App Store.
    2. Mixed payment: Supports USDT/USDC payments. Funds are automatically transferred to the collateral contract.
    3. Logistics integration: The plugin automatically retrieves order logistics numbers and pushes them to the OmniPact predictor. Merchants can withdraw only after the logistics status changes to "signed for" and after an N-day cooling-off period.
- Value: Empowers Web2 merchants to seamlessly access the trillion-dollar cryptocurrency consumer market, while addressing trust and fee pain points in cross-border payments.

3. Supply Chain and Enterprise Applications (For Supply Chain & Enterprise)

- Targeted at: supply chain finance platforms, B2B procurement systems, and financial leasing companies.
- Integration mode: API + IoT deep customization.
- operation flow：
    1. ERP integration: The enterprise's ERP system directly submits bulk purchase guarantees to OmniPact via API.
    2. Hardware Trigger: Integrated with IoT-Anchor SDK. When sensors on the cargo container detect that the goods have arrived at the designated port and the temperature is within limits, the smart contract automatically disburses the funds to the supplier.
- Value: Achieve real-time settlement with zero accounts payable, significantly optimizing supply chain cash flow.

4. 3-Step Quick Start

To minimize integration barriers, we've streamlined the onboarding process:

1. Get credentials: Register in the OmniPact developer console to obtain API_KEY and PROJECT_ID.
2. Sandbox Debugging: Conduct simulated transactions using the OmniPact Testnet environment, where we provide free test tokens (Faucet).

3. Mainnet switch: Change one line of code (Network: Mainnet) to seamlessly switch to production.

2.5 Technical Security and Audit Mechanisms

OmniPact safeguards users 'core assets and business reputation, with security being our top priority in all engineering decisions. We reject the notion of' absolute security 'and instead embrace' multi-layer verification 'and' extremely high attack costs.' To this end, we have established a comprehensive defense system spanning the code, operational, and economic layers.

2.5.1 Smart Contract Defense Architecture

During the contract development phase, we adhere to the industry's most stringent security standards (such as the Checks-Effects-Interactions pattern) and implement multiple layers of defense in the logic layer.

- Reentrancy Guard: All functions involving fund outflows are required to use mutual exclusion locks, eliminating the risk of reentrancy attacks like the DAO incident at its source.
- Multi-Signature & Timelock:
    - The protocol's admin keys are not held by a single private key, but are securely managed in Gnosis Safe multi-signature wallets (e.g., 5/8 signature schemes).
    - All critical protocol parameter changes (e.g., adjusting base rates or upgrading contract logic) are subject to a 48-hour lock-up period. This means any upgrade must be announced on-chain for 48 hours before execution, giving the community sufficient time to review and potentially withdraw funds if malicious proposals are identified.
- Emergency Circuit Breaker (ECB):
    - The protocol is equipped with automated risk control probes. If abnormal fund outflows (e.g., excessive large withdrawals within a short period) or oracle quote flash crashes are detected, the contract will automatically activate "Pause Mode," freezing all non-essential interactions until the security committee investigates.

2.5.2 External Audits and Formal Verification

We introduced third-party independent agencies to conduct "saturation strike" reviews of the code.

- Multi-Round Audits: Prior to mainnet deployment, OmniPact core code must undergo cross-audits by at least three industry-leading security firms (CertiK, Trail of Bits, OpenZeppelin, SlowMist, etc.), with all audit reports and vulnerability fixes publicly disclosed.
- Formal Verification: In addition to manual code review, we employ mathematical methods to formally verify the core fund pool logic. By establishing mathematical models, we demonstrate that the code logic complies with expectations under all possible states, ensuring no logical overflow or deadlock vulnerabilities exist.

2.5.3 Economic Security and Anti-Malice

In addition to code security, we employ game theory design to counteract economic attacks.

- Cost of Attack: In a Distributed Arbitration Network (DAN), an attacker attempting to

manipulate arbitration decisions (51% attack) must acquire and stake over half of the circulating $PACT tokens in the market. As token value increases, the cost of such an attack grows exponentially, far exceeding the potential illicit gains (bribe amount).

- Slashing: When a node is found to be malicious (e.g., a validator submitting falsified data or an arbitrator engaging in vote rigging), the protocol automatically triggers Slashing, confiscating all its staked collateral and transferring the funds to the treasury.

2.5.4 Bug Bounty Program

We believe white hats are the best guardians of the ecosystem.

3. Core Business Operational Mechanics

The OmniPact protocol operates under the rigorous OES (Omni Escrow Standard) state machine logic. We categorize the lifecycle of a standard escrow transaction into four key phases: Agreement, Locking, Fulfillment, and Settlement.

3.1 Full Process Analysis of Standard Escrow Transactions (Standard Escrow Transaction Flow)

Phase 1: Agreement & Initialization

The process begins with consensus. Buyers and sellers negotiate the transaction terms (Term Sheet) off-chain and convert them into on-chain contracts via the OmniPact interface or API.

- Parameter definition: Both parties shall establish core parameters, including: item description (hash and blockchain recording), transaction amount (e.g. 1000 USDT), delivery deadline, and specific acceptance criteria (e.g. logistics confirmation of receipt).
- Contract deployment: Once the buyer confirms the terms, the Factory Contract automatically generates a unique OES guarantee contract instance. At this point, the contract is initialized and not yet funded.

Phase 2: Asset Locking & Custody

This is a crucial step in building trust. Buyers no longer need to 'trust the seller out of thin air,' but instead demonstrate their sincerity by 'staking funds.'

- Funding injection: The buyer transfers the agreed crypto assets (Token/NFT) to the OES contract address.
- Status frozen: The status automatically changes to Locked after the smart contract verifies the funds have been received.
    - Security mechanism: At this point, the funds enter a "quantum superposition state" — neither belonging to the buyer (transferred out) nor the seller (conditions not met). Neither party can unilaterally withdraw the funds; only the code logic can unlock this lock.

Phase 3: Off-chain Fulfillment and On-chain Proof

Under the guarantee of the security of funds, the seller begins to fulfill the obligations in the real world or the digital world.

- Action: The seller ships the goods, submits the code, or completes the service.

- Proof of Delivery: The seller must submit proof of performance to the contract by the deadline.
    - For physical items: submit the logistics tracking number for Omni-Link's automated verification.
    - For digital services: Upload the delivery file to IPFS and write the hash value into the contract.
- Status change: Once submission is confirmed, the contract status changes to Active (or Delivered), triggering a notification to notify the buyer for acceptance.

Phase 4: Confirmation & Atomic Settlement

This is the final stage of the transaction. OmniPact provides two settlement paths, balancing security and efficiency.

- **Path A: Manual Confirmation**
    - After receiving and verifying the goods/services, the buyer clicks 'Confirm Receipt' in the dApp.
    - The smart contract immediately unlocks the funds and transfers the amount (after deducting the minimal protocol fee) in atomic transactions to the seller's wallet. The transaction status changes to Finalized, and both parties mutually verify their credit scores.
- **Path B: Auto-Release Timelock**
    - To prevent buyers from deliberately disappearing or delaying confirmation after receiving goods, the agreement includes a built-in cooling-off period (typically 3-7 days).
    - If the seller has submitted valid proof and the buyer has neither taken any action nor initiated a dispute by the end of the quiet period, the smart contract will be deemed as "accepted" by default, automatically triggering the release of funds. This greatly protects the seller's rights.

- Immunefi's Bounty Program: OmniPact will establish a long-term vulnerability bounty pool on platforms like Immunefi. Researchers who identify critical vulnerabilities (Critical Vulnerability) will receive rewards of up to $1,000,000+. This initiative motivates the world's top security experts to continuously audit protocols.

3.2 The Full Dispute Resolution Lifecycle

In real-world commerce, friction is inevitable—be it product defects, subpar service, or shifting demands. OmniPact overhauls the inefficient human intervention model of centralized platforms, adopting instead a DDRM mechanism. This on-chain judicial framework, grounded in game theory and evidence-based principles, ensures impartial rulings even without a centralized authority.

We standardize the dispute resolution process into four rigorous stages:

Phase 1: Controversy Initiation and Staking

When the buyer and seller cannot reach an agreement through negotiation, either party may trigger the 'confirm button'.

- Trigger condition: The buyer or seller may invoke the raiseDispute() function at any time

before the transaction is finalized.

- Status freeze: The contract status changes from Active to Dispute instantly. The capital pool is fully frozen, and the Auto-Release countdown is paused.
- Anti-Spam Staking: To prevent malicious delays or unreasonable disruptions, the initiating party (and subsequently the responding party) must pay an arbitration fee.
    - Economic model: This fee is typically 3% -5% of the transaction amount.
    - The winning party shall refund the fee in full if the case is won, or pay it to the arbitrator as remuneration if the case is lost. This compels both parties to think twice before initiating a dispute.

Phase 2: Bidirectional Evidence Submission and Evidence Preservation

The system enters the "Evidence Window", which is usually 24-72 hours.

- Evidence Upload: Both parties upload evidence materials (chat records, logistics photos, code comparison reports) via the OmniPact interface. All files are encrypted by the client and stored on IPFS, with only the hash values (Hash) being stored on-chain.
- Privacy Protection: Using asymmetric encryption technology, these evidences are visible only to the transaction parties and the arbitrator to be selected, protecting business privacy from leakage.
- Evidence Curing: Once the window period ends, the evidence submission channel is closed. Neither party may submit or modify materials, ensuring the arbitrator reviews the case based on a fixed set of evidence.

Phase 3: DAN Network Arbitration (Arbitration by DAN)

The protocol activates DAN (Decentralized Arbitration Network) to initiate on-chain adjudication procedures.

- Juror Selection: Using Chainlink's VRF (Verifiable Random Function), three (for small claims) or seven (for large claims) arbitrators are randomly selected from globally trusted nodes to form a' temporary tribunal'.
- Independent adjudication: The arbitrator uses the private key to decrypt the evidence files and independently assesses the case.
- Commit-Reveal Voting (CRV):
    - The arbitrator must submit the voting hash (Commit) within the specified time to choose between supporting the buyer, supporting the seller, or splitting the refund proportionally (e.g., 70%).
    - The voting results are then disclosed during the Reveal phase. This mechanism prevents vote-picking and ensures each arbitrator makes an independent decision.

Phase 4: Execution & Settlement

This is the last mile of justice, which demonstrates the power of "code as law".

- Automated Enforcement: Once voting concludes and a majority consensus is reached, the smart contract acquires full authority to execute the share distribution immediately.
    - Example: The ruling is "Buyer wins." The contract will refund the frozen principal to the buyer's wallet during the specified period and distribute the arbitration fee paid by the losing party to the winning arbitrator.
- SBT Update:
    - The winning party: O-SBT credit score has been improved.
    - The losing party, O-SBT, saw a sharp drop in its credit score and was recorded as

a 'disputed loss' on the blockchain, which will negatively impact its future commercial lending quotas and order acceptance rates.
    - Malicious Arbitrator: If an arbitrator's vote significantly deviates from the final outcome (e.g., by voting against the decision), their staked $PACT tokens will be slashed.

3.2.1 The Ultimate Defense: The High-Level Final Appeal Panel (Supreme Court/Final Appeal Panel)

To address rare cases involving complex, highly contentious, or potentially systematic attacks, the OmniPact protocol activates the highest judicial level—Supreme Court Intervention—when conventional DAN random arbitration and appeal procedures fail to deliver a convincing fair outcome.

1. The Role of the Final Review Panel (Positioning)

The final panel is not a random algorithm-selected "civilian jury", but a "senate" composed of elite nodes with the deepest interest ties and the highest credibility in the ecosystem. Their rulings are final and irrevocable, directly determining the allocation of funds, and no appeals are accepted.

2. Strict Selection Criteria

Supreme Jurors must satisfy two stringent criteria to ensure they possess sufficient wisdom to distinguish right from wrong and the financial capacity to bear the consequences of misjudgment.

- High O-SBT Score:
    - A user's Omni-Score must rank in the top 1% across the entire network (e.g.,>900 points).
    - They must hold the 'Senior Arbitrator' badge and demonstrate a historical decision coherence rate exceeding 98%, signifying their status as seasoned 'wise' arbitrators.
- High Stake (Skin in the Game):
    - The agreement must include a substantial $PACT stake (e.g., over $500,000).
    - This deposit functions as a 'credit guarantee.' Should the final adjudication panel be found guilty or issue a verdict that contradicts basic facts, the substantial sum will be forfeited (Slash), with the cost far exceeding any potential bribe.

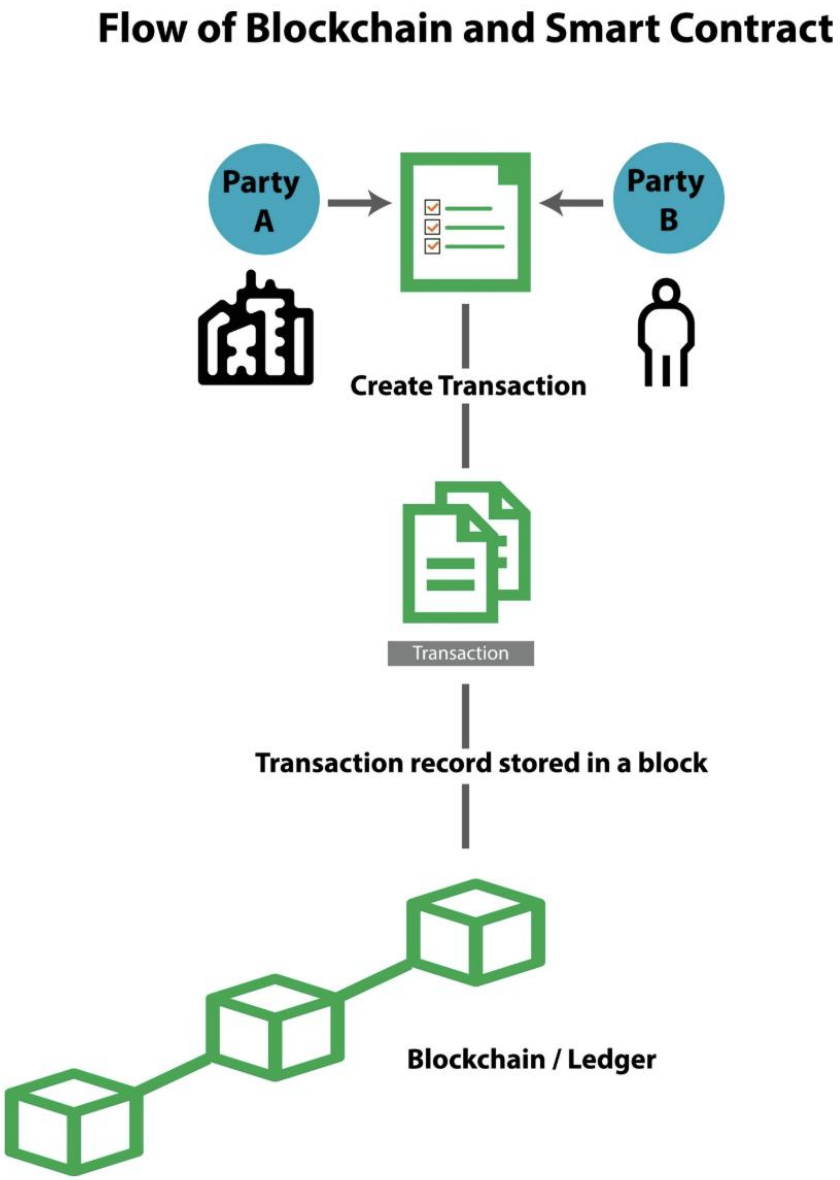
3. Triggering Mechanism

The Final Panel will not readily intervene in ordinary disputes, but will only do so if the following conditions are met:

- Exhaustion of appeal stages: When a case has undergone multiple rounds (e.g., three) of appeals through the standard DAN network, with both parties remaining dissatisfied and the disputed amount being substantial.
- The amount of pledge is capped: when the arbitration fee paid by the appellant reaches more than 50% of the amount of dispute, it shows that the party thinks that they have suffered great injustice and are willing to pay a huge price to seek the truth.
- Community Emergency Proposal: Should the community identify clear signs of 'governance attacks' in a routine ruling, it may propose through the DAO to urgently suspend its enforcement and refer the matter to the Final Review Panel for reassessment.

4. The Ruling Process

- Open Identity Voting: Unlike the anonymous blind voting in standard arbitration, the final panel may cast votes under real names (linked to their Omni-ID).
    - Logic: Elite members must validate their long-term reputation for the judgment. Through public voting records, they are subject to supervision by the entire community.
- Supermajority rule: Final rulings typically require 2/3 (66%) or even 3/4 (75%) of the votes to be weighted in favor to take effect, rather than the simple majority (51%).
- Final execution: Once the decision is generated, the smart contract immediately executes the fund distribution. Any subsequent objections can only be resolved through off-chain legal channels, and the on-chain state is irreversible.



3.3 Cross-Chain Escrow Transaction Mechanism

The future of Web3.0 lies in interconnected multi-chain ecosystems. To break down liquidity silos, OmniPact developed the X-Escrow (Cross-Chain Escrow) mechanism, enabling users across diverse blockchain networks (including Ethereum, BNB Chain, Solana, and Layer 2 networks) to execute collateralized transactions with the same seamless efficiency as on-chain operations.

We employ a dual-layer architecture combining state synchronization and liquidity aggregation to ensure the atomicity and security of cross-chain transactions.

1. Core Logic of Cross-Chain Transactions: Chain-Agnostic Settlement

OmniPact decouples 'funding custody' from 'business logic' by transmitting state signals through underlying interoperability protocols like Chainlink CCIP or LayerZero.

- Scenario: Buyer Alice holds USDC on Arbitrum, while seller Bob wants to receive USDT on BNB Chain.
- Traditional pain points: Alice must first cross-chain independently, exchange tokens, pay high Gas fees, and then initiate collateral on BNB Chain. The process is cumbersome and high-risk.
- OmniPact solution: Alice makes one-click payments directly on Arbitrum, with the system automatically handling all subsequent operations.

2. X-Escrow Operational Workflow

Step 1: Source Chain Locking

- The buyer initiates the transaction on Chain A.
- Smart contracts securely hold buyers' funds (including the transaction principal and estimated cross-chain Gas fees) in the OmniPact source chain vault.
- The contract triggers a CrossChainLock event containing the target chain ID, seller address, and transaction terms hash.

Step 2: State Relay & Verification

- The decentralized relayer network (Relayer Network) captured this event.
- After passing multiple security verifications (DON Consensus), a cross-chain message containing a Merkle Proof is generated and sent to the target chain.

Step 3: Destination Chain Mirroring

- The OmniPact contract on Chain B receives and verifies messages.
- The contract generates a 'Shadow Escrow' locally.
- After seeing the status change to 'Locked (Verified)' on Chain B, seller Bob confirmed the funds were securely locked on Chain A and proceeded to ship or fulfill the order.

Step 4: Atomic Settlement & Release

- When the transaction is completed (buyer confirmation or arbitration approval):
    - Command back: Chain B sends a 'release' signal to Chain A.
    - Liquidity routing: Chain A's vault releases funds, which are automatically converted to USDT via integrated aggregators (e.g., 1inch) and transferred cross-chain to Bob's wallet on Chain B; alternatively, Bob may directly receive the assets on Chain A if desired.
    - Rollback on failure: If the transaction is canceled or the arbitrator rules for a refund, the cross-chain message will trigger Chain A to release the funds and return them to Alice. The entire process is either fully successful or fully rolled back, with no risk of losing funds in an intermediate state.

3. Key Technical Features

- Gas Abstraction: Users don't need to hold native tokens (e.g., BNB) from the target chain to pay Gas fees. OmniPact automatically deducts and pays all Gas fees for cross-chain transactions from the stablecoins on the source chain.
- Unified Liquidity Pools: OmniPact maintains liquidity pools on mainstream blockchains. For frequently traded assets like USDT/USDC, it employs the 'Lock & Mint' or 'Liquidity Rebalancing' mechanism, enabling instant deposits without the lengthy native bridge confirmation process.

3.4 Batch Transactions and Conditional Airdrop Escrow

In the Web3 ecosystem, project teams frequently distribute rewards to thousands of users—such as marketing airdrops, bug bounty programs, or DAO contributor compensation. The traditional centralized distribution model has two major drawbacks: first, users cannot verify if the project team actually has the funds to pay out; second, the project team might default on payments after users complete tasks.

OmniPact introduced the Batch-Escrow mechanism, which establishes a zero-trust marketing and compensation infrastructure through a combination of 'pre-locked fund pools' and 'conditional automated distribution'.

1. Core Logic: The Dispenser Contract

Unlike one-to-one OES contracts, batch guarantees operate on a 'fund pool' model.

- Proof of Funds: The sponsor (e.g., the project team) must first deposit the total budget (e.g., 100,000 USDT) into the OmniPact distributor contract in a single transaction.
- Irrevocable period: After funds are locked, a "task window period" is set. During this period, the initiator cannot withdraw the funds at will, ensuring that the user's efforts (such as retweeting tweets or testing code) are fully compensated.

2. Conditional Triggering and Verification

How to determine who qualifies for the capital pool? We use Omni-Link Prophet for automated verification.

- Social verification: For marketing campaigns, integrate Twitter/Discord API. When the oracle verifies the user has completed a specified action (e.g., Retweet ID: 12345), the user's address is automatically added to the whitelist.
- On-chain behavior verification: For interaction rewards, the contract automatically checks users 'interaction records on specific protocols (e.g.,' 5 transactions on Uniswap') and awards rewards upon meeting the criteria.
- DAO contribution verification: For salary distribution, the voting results are processed through the multi-signature wallet (Gnosis Safe). Once the DAO vote approves the monthly payroll, OmniPact automatically executes batch transfers.

3. High-efficiency distribution mechanism: Merkle Tree and Pull model (Gas Efficiency)

To save Gas fees for thousands of transfers, we offer two distribution modes:

- Push mode (active push-suitable for up to 100 people):
    - Suitable for DAO core contributors' salaries. The smart contract automatically transfers funds to multiple addresses in a single transaction (Batch Transfer).
- Pull mode (Merkel collection-applicable to 10,000+ people):
    - Designed for large-scale airdrops. The protocol calculates and records the Merkle

Tree Root for all eligible users on the blockchain.
    - When users click 'Claim' on the frontend, they submit a Merkle Proof. Upon successful verification, they can withdraw funds independently. This method transfers Gas fees to the recipient, significantly reducing the project's financial burden.

4. Refund Mechanism

If there is a balance in the fund pool after the event (for example, the original reward was for 1000 people, but only 800 actually completed the task):

- Auto-refund: After expiration_time, the contract automatically unlocks the remaining funds and refunds them to the initiator.
- Burn/Donation: The initiator may also choose to automatically burn the remaining funds or donate them to the OmniPact treasury upon creation, demonstrating their commitment to deflationary principles.

3.5 Reputation Scoring and Application Mechanism

In the OmniPact ecosystem, reputation is regarded as a more critical trust asset than capital. We developed the Omni-Score dynamic scoring algorithm, which converts every user interaction, fulfillment, and dispute record on the chain into a quantifiable score (0-1000 points). This score is stored in the user's SBT, serving not only as an identity marker but also as the core parameter determining their economic privileges.

1. Omni-Score Scoring Algorithm Model (The Scoring Algorithm)

Omni-Score isn't just a simple linear adder—it's a multi-dimensional weighted model with a time decay factor. Its core logic rewards consistent honesty while delivering devastating consequences to malicious behavior.

Score Formula Logic Overview:

$$R_{total} = (W_p \cdot P + W_v \cdot V + W_s \cdot S) \cdot T - D$$

The parameters for each dimension are defined as follows:

- P (Performance-Performance):
    - Key metric: Transaction success rate (number of non-disputed transactions / total transactions).
    - Weight: highest (approx. 50%). This is the most direct measure of adherence.
- V (Volume-Asset Depreciation):
    - Key metric: Total Value Locked (TVL).
    - Weight: Medium (approximately 30%). Prevents new accounts from artificially inflating scores and establishes the weight of high-value users, but imposes a logarithmic cap (Log Cap) to prevent whale monopolization.
- S (Social-Social Verification):
    - Key metrics: User reviews from other high-reputation users (Web of Trust) or verified DAO membership.
    - Weight: Low (about 20%).
- T (Time Decay-Time Decay Factor):
    - Logic: Reputation needs to be "fresh". The T value decreases over time. Users must remain active (with recent transactions) to maintain a high score.

Performance records from 3 years ago will be significantly devalued.

- D (Dispute Penalty-Dispute Penalty):
  - Logic: Once fraud is determined in arbitration (losing the case), points will be deducted exponentially. One serious offense may negate 100 past honest accumulations, even leading to zero credibility.

2. Reputation Tiers & Economic Privileges

Based on the Omni-Score, users are categorized into 5 tiers. The higher the tier, the lower the protocol cost and the greater the permissions. This creates a positive flywheel where earning money becomes easier with higher credit.

Tier	Score	fee discount	Key Privileges
Lv 1	0 - 200	0% (full)	Only basic guarantee features are available. A full deposit is required.
Lv 2	201 - 500	10% OFF	Unlock high-frequency API call permissions.
Lv 3	501 - 700	30% OFF	Credit guarantee: allows a 90% collateral ratio (under-collateralized).
Lv 4	701 - 900	50% OFF	Arbitrator Qualification: Eligible to be elected to DAN and receive arbitration fees.
Lv 5	901 - 1000	Free	VIP Access: Exclusive DAO customer support and doubled voting power for DAO governance proposals.

3. Credit Scenarios (Application Scenarios)

A. Dynamic Collateralization

Traditional DeFi lending typically requires high collateral (e.g., 150%). In OmniPact, however, highly reputable merchants can access low-collateral guarantees.

- Scenario: A seasoned Level 4 OTC dealer initiates a sell order with just 20% margin collateral (backed by credit), significantly boosting capital efficiency.

B. Juror Eligibility

The fairness of the DAN network depends on the integrity of its arbitrators. Only users with an Omni-Score above 700 are eligible to stake $PACT to nominate arbitrators. This means the arbitrators themselves must be proven honest over time.

C. Priority Matching and Display

In third-party platforms (e.g., freelance marketplaces) that integrate the OmniPact SDK, Omni-Score is converted into 'recommendation weight'.

- Scenario: When buyers search for 'Logo Design,' designers with Level 5 credibility are prioritized and displayed with the OmniPact Certified' gold shield badge, thereby securing more orders.

4. Contract Guarantee Panorama & Ecosystem

OmniPact core vision is "Guarantee Everything". Leveraging OES standards and the Omni-Link oracle network, we transform complex business transaction logic into configurable smart contract templates.

Here are the 10 core guarantee categories natively supported by OmniPact, spanning from purely on-chain digital interactions to complex real-world economic activities.

4.1 Panoramic View of Guarantee Categories

Field 1: Digital Native Services

Such scenarios utilize off-chain data oracle and file hash evidence to achieve automated or semi-automated fulfillment verification.

1. Technical & Development Escrow

- The pain point: the common risk of 'unfinished projects' in outsourcing development; employers refusing to pay after developers deliver code.
- OES mechanism: Milestone-Based Release.
    - The agreement integrates with GitHub/GitLab API. Funds are locked in phases (e.g., 30% deposit, 30% Alpha version, 40% delivery).
    - Trigger conditions: The contract automatically releases corresponding phase funds when the developer submits a PR (Pull Request) and passes CI/CD automated testing, or after multi-signer approval by the employer.

2. API Interface and Service Guarantee (API & SaaS SLA Guarantee)

- Key pain points: SaaS service provider outages, API response delays exceeding standards, and user prepaid funds being compromised.
- OES mechanism: SLA performance-based staking (Performance-Based Staking).
    - Trigger conditions: Uptime and Latency of Omni-Link real-time monitoring service provider. If the monitoring data falls below the SLA commitment (e.g., Uptime <99.9%), the contract automatically triggers compensation, refunding part of the prepaid funds to the subscriber.

3. Data Governance and Compliance Safeguards (Data Sovereignty & Compliance)

- Key pain points: Fraud risks in AI training data transactions and privacy breaches after data transactions.
- OES mechanism: TEE (Compute-to-Data) Privacy Computing Sandbox.
    - After the buyer's funds are locked, the data is not sent directly to the buyer but enters the Trusted Execution Environment (TEE) for model training.
    - Trigger condition: When TEE returns the 'Training completed' signal, funds are released to the data seller, while the buyer only receives model weights. The original data remains in the repository, ensuring the seller's data sovereignty.

Domain 2: Asset Management & Distribution

This scenario utilizes on-chain atomic exchange and batch processing capabilities to resolve trust issues in asset transfers.

4. Airdrop Activity Guarantee (1-to-N Airdrop Guarantee)

- Key pain points: Project teams promise 'task-based airdrops' only to disappear (Rug Pull), while users fall victim to volume fraud.

- OES mechanism: Pre-funded Pool.
    - The project owner must first inject the total prize pool into the contract.
    - Trigger condition: When the oracle verifies a user's social behavior (e.g., retweeting) or on-chain interactions meet the criteria, the user's address is automatically added to the Merkle Tree whitelist, enabling them to claim rewards independently and preventing project teams from defaulting.

5. Information and Digital Assets Trading Guarantees (Digital Assets & Accounts)

- The pain point: the risk of account recovery in gaming accounts, social media influencers, and domain name trading.
- OES mechanism: Ownership Transfer Verification + Cooling-off.
    - Trigger conditions: For assets supported by API queries (e.g., GoDaddy domains, Steam accounts), the contract verifies ownership change records. For unsupported assets, a 7-14 day cooling-off period is set. If the buyer does not file a "account recovered" appeal within this period, the funds will be released automatically.

6. End-to-end Intellectual Property and Copyright Lifecycle Protection

- The pain points are: the copyright licensing fees for music and pictures are not paid; and the piracy is difficult to trace.
- OES mechanism: IP-NFT and Royalty Streaming.
    - Convert IP to Omni-NFT. The revenue of the licensee (e.g., peripheral manufacturers) is linked to the revenue-sharing contract.
    - Trigger condition: For each sale generated, the smart contract automatically deducts x% royalty to IP creators, enabling real-time revenue distribution.

7. Parametric Insurance Risk

- Key pain points: Traditional insurance claims processes are cumbersome, and insurers often refuse payouts with fabricated excuses.
- OES mechanism: Parametric Trigger.
    - The underwriter pledges the full amount of the funds.
    - Trigger condition: When the oracle detects objective indicators (e.g., flight delays exceeding 4 hours, local rainfall below the threshold, or DeFi protocol hacks), the contract automatically compensates the policyholder without manual review.

Field 3: Real Economy and RWA (Real World Economy)

This scenario utilizes IoT and logistics oracles to connect the physical world.

8. Third-party dependencies and supply chain guarantees (Supply Chain Dependencies)

- The pain point is the multi-tier supplier triangle debt, and the upstream supply cut-off leads to the downstream default.
- OES mechanism: Chained Splitting.
    - Trigger condition: Upon end customer confirmation of delivery, the smart contract automatically distributes funds proportionally to tier-1 suppliers, tier-2 raw material suppliers, and logistics providers in a domino effect, effectively resolving supply chain payment delays.

9. Real World Assets (RWA) tokenization collateral

- Key concern: Are the on-chain tokens linked to real properties/vehicles actually existing? Are they subject to duplicate mortgages?

- OES mechanism: IoT-Anchor proof of existence.
    - Trigger condition: IoT devices on leased assets (e.g., shared cars) must send heartbeat signatures regularly. If the device goes offline or the sensor triggers an alarm (e.g., being dismantled), the contract will automatically suspend rental income payments to the asset owner and initiate liquidation.

10. Physical Goods Fulfillment

- Key pain points in cross-border e-commerce include fake shipments, mismatched products, and lost items during logistics.
- OES mechanism: Logistics Hook.
    - Trigger condition: Integrate DHL/FedEx global logistics data. Funds remain locked until the logistics status changes to Delivered (signed for). If a package is lost (status: Lost), the contract automatically initiates a claim or refund process.

4.2 Core Functional Features

OmniPact role as the universal trust layer in Web3.0 stems from the deep integration of its four core functional features. These features collectively form a self-contained, evolving, and boundaryless credit ecosystem.

1. Full-Process Automated Execution

— Eliminating human friction with code (Efficiency)

Traditional guarantees rely on manual verification processes (e.g., bank manager signatures or platform customer service order checks), which are inefficient and error-prone. OmniPact enables deterministic execution throughout the entire transaction lifecycle.

- Atomic Settlement: Under the OES standard, the protocol binds 'payment' and 'delivery verification' into a single blockchain transaction. The process is either fully successful or completely failed, eliminating any intermediate state like 'payment received but goods not delivered.'
- 7x24h Smart Trigger: Real-time smart contract responses without downtime. Whether it's a logistics confirmation at 3 AM or weekend code submission, funds will be released within seconds once Omni-Link verifies the conditions.
- Zero-Touch: In 95% of normal transactions (Happy Path), no third-party intervention is required from creation to completion, significantly reducing operational and time costs.

2. Multi-Chain Asset Interoperability

— Breaking the liquidity silos (Reach)

OmniPact is not merely a DApp deployed on a single chain, but a cross-chain universal protocol (Omni-Chain Protocol).

- Chain-Agnostic Payment: Buyers can use A-chain assets (e.g., Arbitrum USDC) to purchase services or goods on B-chain. The protocol's X-Escrow module automatically handles cross-chain messaging and fund locking.
- Gas Abstraction: Users don't need to hold native tokens (e.g., ETH or BNB) of the target chain to pay Gas fees. OmniPact smart contracts automatically deduct a minimal fee from the transaction amount and cover it, delivering a web2-like seamless experience.
- Unified liquidity routing: By integrating leading cross-chain bridges (e.g., LayerZero,

Wormhole) and DEX aggregators, it ensures optimal exchange rates and minimal slippage for funds during transactions.

3. Decentralized Dispute Resolution

-Algorithmic Justice (Fairness)

When automated processes encounter subjective disagreements, OmniPact provides not a 'customer service ruling,' but a ** 'judicial ruling.' **

- Anti-Review and Anti-Bribery: The DAN network, designed based on the Schelling Point Game, employs random sampling (VRF) and blind voting mechanisms (Commit-Reveal) to prevent attackers from predicting or bribing arbitrators.
- Unalterable evidence: All disputed evidence (chat logs, files) is stored via hash anchoring on IPFS/Arweave. Even protocol developers cannot delete or modify evidence favorable to users.
- Community Jury System: Decisions are returned to the community. Highly credible arbitrators from diverse global backgrounds leverage their expertise (e.g., code audit experts resolving development disputes) to deliver more professional judgments than general customer service.

4. Composable On-Chain Reputation

— Visualizing and Capitalizing Trust (Value)

The biggest innovation of OmniPact lies in transforming "transactional behavior" into "credibility assets". The SBT-based Omni-ID system has high composability, becoming a universal "credit Lego" in the Web3 ecosystem.

- Cross-platform interoperability: Omni-ID's credit data is open.
    - Loan Agreement: Accesses users' Omni-Score to provide low-collateral loans to high-scoring users.
    - DAO organization: automatically distributes governance airdrops or whitelists to high-reputation users.
    - HR platform: Recruitment websites can leverage users' performance records as professional credentials.
- Verifiability of privacy protection: By leveraging zk-SNARKs (zero-knowledge proofs), users can demonstrate their creditworthiness to third parties without disclosing sensitive transaction details such as amounts or counterparties.
- Positive incentive flywheel: By valuing reputation (e.g., fee reductions and higher order acceptance rates), users voluntarily uphold transaction integrity, thereby reducing the network's overall regulatory costs.

4.3 Product Iteration Roadmap

While OmniPact's vision is ambitious, our execution is pragmatic and agile. We have structured the protocol's development into four strategic phases, progressively building a complete closed loop from the foundational protocol to a global ecosystem.

Phase 1: Genesis-The Protocol Kernel and Cold Start

Objective: Complete the development of the core protocol layer, validate the feasibility of the OES standard, and ensure the secure implementation from scratch.

- **Q1-Protocol Definition and Development**

- Launch the OmniPact White Paper V1.0.
- Complete the architectural design and coding of the core smart contract for OES (Omni Escrow Standard).
- Develop the Alpha version of the OmniPact dApp (Web version), supporting basic ERC-20 token-guaranteed transactions.

- **Q2 – Safety Audit and Testing Network**
  - Launch the Testnet and initiate the 'Bug Bounty Hunting' program.
  - Complete the initial external code audit (performed by CertiK or SlowMist).
  - The OmniPact SDK V1.0 (Beta) has been released, inviting the first 10 Web3 developers to participate in the private beta integration.
- **Q3-Mainnet Launch**
  - Deploy the protocol to Ethereum and BNB Chain mainnet.
  - The 'Genesis Guarantee Mining' campaign has officially launched to attract early liquidity.
  - Deploy the OmniPact private network.
  - The SBT Basic Credit System has been launched, supporting on-chain recording of basic compliance data.

Phase 2: Evolution-The Arbitration Network and Economic Closed Loop

Objective: To implement decentralized governance and value capture of the protocol by introducing the DAN arbitration network and token economy model.

- **Q4-DAN Network Pilot**
  - DAN (Decentralized Arbitration Network) module is released.
  - The first 100 Genesis Jurors are recruited to participate in a mock trial.
  - With Omni-Link Prophet integrated and mainstream logistics APIs (DHL/FedEx) connected, the e-commerce fulfillment loop is seamlessly closed.
- **Year 2 Q1-TGE and Token Ecosystem**
  - $PACT Token Generation Event (TGE): Official token issuance and listing on the exchange.
  - Launch the $PACT staking system, enabling public selection of arbitrators and node staking.
  - The fee discount mechanism has been implemented, with SBT's credit score now officially linked to economic benefits.
- **Q2 of Year 2-Mobile and Service Layer Expansion**
  - The OmniPact iOS/Android app is now available, featuring one-click mobile collateralization and push notifications.
  - The integrated IPFS/Arweave decentralized evidence system supports encrypted upload of large file evidence.

Phase 3: Expansion — Cross-chain Interoperability and Real-World Application Deployment

Objective: To break down the barriers between chains, bridge the virtual and the real, and

expand B2B commercial scenarios on a large scale.

- **Year 2 Q3-X-Escrow Full Chain Interoperability**
    - The integrated Chainlink CCIP/LayerZero platform has launched its X-Escrow cross-chain escrow module.
    - Enable 'Pay with Any Token' to seamlessly integrate with major ecosystems including Ethereum, Solana, and Arbitrum.
    - Launch Shopify/WordPress payment plugins to officially enter the Web2 e-commerce market.
- **Year 2 Q4-RWA and Internet of Things Integration**
    - The IoT-Anchor SDK is now available, enabling direct integration with smart devices such as smart locks and connected vehicle terminals.
    - Launch the first RWA benchmark case (e.g., pilot programs for decentralized housing rental or supply chain finance).
    - Launch the Global Eco-Partnership Program to build localized communities in cross-border trade hotspots like Southeast Asia and Latin America.

Phase 4: Ubiquity (Boundless Era) — AI Integration and Full Autonomy

Objective: To establish a globally interoperable trust infrastructure that enables full DAO takeover.

- **Year 3+ -Privacy Matrix and AI Arbitration**
    - We have fully deployed the Privacy Matrix and launched zk-SNARKs privacy verification capabilities to serve enterprise-level clients.
    - Introducing AI-assisted jurors (AI-Juror) to leverage large language models, enabling human arbitrators to rapidly analyze evidence and enhance trial efficiency.
- **The Final Form-DAO Ultimate Governance**
    - Remove all administrator private keys (Admin Keys Burned) and transfer full protocol parameter adjustment authority to OmniPact DAO.
    - The protocol achieves self-sustaining and self-evolving capabilities, evolving into a perpetually operational public trust infrastructure in the Web3 ecosystem.

5. Tokenomics Model

The $PACT token is a native functional and governance token in the OmniPact Protocol ecosystem. It not only serves as a medium for network value transfer but also acts as the core incentive mechanism to ensure security and consensus in the Decentralized Arbitration Network (DAN).

5.1 Token Basic Information

$PACT's design is based on the principles of fixed total volume, deflationary model, and multi-chain interoperability.

1. Core Parameter Specifications (Key Specifications)

attribute (Attribute)	Specifications	Description
Token full name	**OmniPact Token**	The official entitlement certificate.
Ticker	**$PACT**	PACT/USDT exchange code.
Total Supply	**1,000,000,000**	1 billion (fixed total, never increased).
accuracy (Decimals)	**18**	Complies with ERC-20 standard precision and supports micropayments.
contract standard	**OFT (Omni-Chain Fungible Token)**	Native cross-chain transfer is supported based on the LayerZero/CCIP standard.
initial offering network	**Ethereum (Mainnet)**	Core assets settlement layer.
Support network	BNB Chain, Solana, Arbitrum, Optimism	Implement 1:1 mapping through the official bridge contract.

2. Contract Features

To address complex business and governance needs, $PACT smart contracts incorporate the following advanced features:

- Native Cross-Chain:
  - $PACT operates on the OFT (Omni-Chain Fungible Token) standard. This means tokens are transferred between chains via a contract-level Burn-and-Mint mechanism rather than through vulnerable 'lock-and-mint' cross-chain bridges, eliminating the risk of asset de-anchoring due to traditional cross-chain bridge hacks.
- Governance Voting:
  - The token features built-in Delegate and Vote functions, compatible with off-chain governance tools like Snapshot and On-chain Governor contracts.
- Security Pause:
  - In the event of extreme security incidents, the DAO committee may activate a temporary transfer suspension to safeguard token holders' assets.
- Flash Loan Protection:
  - The governance voting and arbitration staking module incorporates a built-in Snapshot Checkpoints mechanism to prevent attackers from leveraging lightning loans to borrow massive amounts of $PACT within a single block, thereby manipulating voting outcomes.

3. Value Capture Logic Overview

$PACT is more than just a governance token—it captures the economic value of the entire protocol network.

- As fuel: payment of arbitration fees and senior agreement service fees.
- As a 'work credential': You must stake $PACT to become a Juror and earn rewards.
- As a 'deflationary asset', a portion of the agreement revenue will be used to repurchase and destroy $PACT.

5.2 Core Token Utility

$PACT tokens serve as the lifeblood of the OmniPact protocol's economic ecosystem. More than just a payment tool, they are a multifunctional asset that integrates governance, yield, and work access. Through a meticulously designed utility model, $PACT captures the growth value of the entire staking network.

1. Decentralized Governance

— Holding tokens grants authority. As the protocol evolves into a DAO (Decentralized Autonomous Organization), $PACT holders will become the ultimate decision-makers.

- OIP Voting: Holders can vote on OmniPact Improvement Proposals.
    - Governance scope: Adjusting base agreement rates (e.g., from 0.5% to 0.3%), modifying arbitrator collateral thresholds, adding or removing supported asset classes, and even directing the allocation of treasury funds.
- Parameter adjustment authority: Through on-chain governance contracts, the community can directly modify key parameters, ensuring the protocol evolves flexibly based on market feedback rather than being dictated by the founding team.

2. Arbitration Incentives & Work Staking

— The gateway to justice. This is $PACT primary demand source. Within OmniPact DAN network, $PACT functions as a Work Token.

- Juror Staking: To become a juror and earn arbitration fees, users must stake a specified amount of $PACT in the protocol.
    - The higher the collateral volume, the greater the probability of being selected to handle cases, and the higher the expected annual percentage yield (APY).
- Rewards & Slashing:
    - Reward: Arbitrators who make a fair decision (in line with the majority) will receive an arbitration fee reward paid in $PACT.
    - Penalty: If an arbitrator is found guilty of misconduct (e.g., voting against the facts), their staked $PACT will be slashed and destroyed. This mechanism ensures alignment between cybersecurity maintenance and token value preservation.

3. Transaction Fee Discounts & Burn

— Reducing user costs and creating deflation. Much like BNB role in Binance's ecosystem, $PACT provides direct economic incentives for high-frequency traders.

- Discount on payments: When creating a trust-based contract (guaranteed contract), users who choose to use the $PACT payment protocol service fee can enjoy a discount of 20% to 50% (the discount rate decreases annually).
    - Effect: This incentivizes all OmniPact users and individual traders to buy and hold $PACT in the market.
- Position Waiver: The system automatically determines the fee tier (Tier) based on the $PACT holdings in the user's wallet and SBT credit rating via Oracle. Large accounts qualify for zero-fee privileges.

4. Staking and Insurance Fund Participation

— Security Cushion for Protocols To address extreme black swan events (such as financial losses

caused by vulnerabilities in smart contracts), the protocol incorporates a Safety Module.

- Insurance Mining: Holders can stake $PACT in the insurance fund pool.
    - Risk: In the event of a payment default under the agreement, the pool funds will be auctioned to compensate affected users.
    - Income: As compensation for assuming risks, the pledgor will receive 30% of the protocol treasury revenue (Real Yield, distributed in USDT/ETH) plus additional $PACT release rewards.

5. Ecosystem Authorization & Payment

— The universal currency for value-added services. In OmniPact expansive ecosystem, $PACT serves as the exclusive settlement currency.

- Advanced API Access: Third-party integrators (e.g., exchanges, SaaS platforms) must purchase $PACT to obtain API Credits for accessing high-frequency APIs or accessing advanced risk control data.
- Advertising and Promotion: Premium merchants seeking Featured placements on OmniPact dApp's homepage or search results must bid to redeem $PACT.
- RWA device access fee: IoT device manufacturers must pay a nominal $PACT fee as a registration fee when connecting to the OmniPact network.

5.3 Token Allocation Scheme and Vesting Schedule

The $PACT allocation model balances early adopter incentives, the protocol's long-term funding needs, and the community's decentralized governance goals.

5.3.1 Allocation Breakdown

For the specific plan regarding token issuance and distribution, please refer to the official OmniPact documentation.
https://docs.omnipact.io/about/token-economic-model/token

5.4 Incentive Gradient Details

The OmniPact economic model is built on the core principle that 'contribution determines returns.' To avoid egalitarianism, we've developed a sophisticated multi-dimensional incentive gradient system. This system guides ecosystem participants to evolve from mere' speculators 'into long-term' builders' through tiered differences in three dimensions: rates, returns, and governance rights.

1. Transaction Fee Discount Gradient

— Reduce friction costs for high-frequency users

The basic service fee for the protocol is set at 0.5% -1% of the transaction amount. To incentivize users to hold $PACT and accumulate on-chain reputation, we implement a "dual-factor discount model": the discount level depends on the user's $PACT holdings and Omni-ID reputation tier.

Tier	position requirement	Creditworthiness Requirements (Omni-Score)	rate discount	Payment method bonus
Tier 0	0	< 200	0% (original price)	not have

Tier	position requirement	Creditworthiness Requirements (Omni-Score)	rate discount	Payment method bonus
Tier 1	> 1,000	> 200	**10% OFF**	Apply an additional 5% discount using the $PACT payment rate
Tier 2	> 10,000	> 500	**20% OFF**	Apply an additional 10% discount if using the $PACT payment rate
Tier 3	> 50,000	> 700	**40% OFF**	Apply an additional 15% discount using the $PACT payment rate
Tier 4 (VIP)	> 200,000	> 900	**80% - 100%OFF**	Free

Note: B-side access providers (e.g., e-commerce platforms) with millions of users can apply for a permanent API royalty-free channel by destroying a fixed quantity of $PACT.

2. Arbitration Incentive Gradient

— Rewarding professionalism and fairness

The DAN network is not a simple 'headcount voting' system, but rather an elite governance framework based on weighting. Arbitrators' Yield is closely tied to their staking volume and **historical performance (Coherence)**.

- Base Income Formula:

$$E = (F_{arb} + I_{inflation}) \times W_{stake} \times M_{reputation}$$

  - F_{arb}: The arbitration fee paid by both parties in the case.
  - $I_{inflation}$: Token rewards from protocol issuance (early-stage subsidies).
  - W_{stake}: Staking weight. The more you stake, the higher your chance of being selected.
  - $M_{reputation}$: Reputation multiplier.
- Reputation Multiplier:
  - Novice arbitrator (accuracy <80%): multiplier is 0.8x (discounted returns).
  - Senior arbitrator (accuracy> 95%): multiplier is 1.5x (excess return).
  - Expert arbitrator (50 consecutive wins): multiplier is 2.0x.
- Punishment gradient for evil deeds:
  - First misjudgment: deduct the current income.
  - Second misjudgment: deduct 10% of the pledged principal.
  - Malicious collusion/accepting bribes: Slash 100% of the principal and permanently disqualify.

3. Staking Yield & Time-Lock

— Rewarding Long-Term Investors

To mitigate selling pressure in the circulating supply, OmniPact adopted a veToken (Vote-Escrowed) model modeled after Curve. Users staking $PACT receive non-transferable vePACT stake instead of standardized returns.

- Time-Lock Multiplier:
  - Lock-in for 1 month: Earn 1x vePACT weight.

    - Lock-in for 6 months: Earn 2x vePACT weight.
    - Lock-in period: 4 years (maximum). Earn 10x vePACT.
- Real Yield:
    - 50% of the protocol's treasury revenue (calculated in ETH/USDT) will be distributed weekly to vePACT holders.
    - This means that the longer you hold your positions, the more 'cash flow' you receive—potentially exceeding your initial investment.

4. Governance Power Gradient

— Avoiding a Cetacean Dictatorship

DAO governance is directly tied to the vePACT count rather than the $PACT balance. This ensures that only users aligned with the protocol's long-term interests can participate in decision-making.

- Proposal Threshold: Only addresses (or address sets) holding 1% of vePACT across the network are eligible to propose OIP governance.
- Quadratic Voting: This mechanism applies to specific livelihood issues (e.g., fee adjustments) through the formula $Cost = Votes^2$, reducing the voting dominance of super whales and safeguarding the voices of small and medium-sized businesses.
- Veto Power: The Guardian Council, comprising core contributors and top-tier trusted nodes, holds emergency veto power to block proposals during governance attacks.

5.5 Token Burn Mechanism

To ensure the long-term growth of $PACT value, OmniPact has implemented a deflationary model dubbed "The Black Hole". We pledge to allocate a significant portion of the protocol's captured economic value to repurchase and permanently burn $PACT until the total supply drops to 12% of its initial volume (120 million coins).

This mechanism ensures that every collateralized transaction on the OmniPact network increases the scarcity of $PACT at the micro level.

1. Buyback & Burn Engine

— Convert agreement revenue into token buy orders

The service fee charged by the OmniPact protocol (typically 0.5%-1% of the transaction value) is primarily held in the treasury in the form of mainstream assets like USDT, USDC, and ETH. We will enable smart contracts to automatically execute repurchase orders.

- Income distribution: 20% to 40% of the protocol income (determined by DAO vote for quarterly allocation) will be allocated to the 'stake pool'.
- Automated execution:
    - Smart contracts execute market buy orders on a regular basis (e.g., weekly) on decentralized exchanges (DEX) such as Uniswap and PancakeSwap.
    - The purchased $PACT tokens do not enter any wallet, but are sent directly to the Null Address (0x00...dead) in an atomic transaction.
- On-chain audit: Every Tx Hash repurchased and destroyed is published on the official website and can be verified by anyone, eliminating the "fake destruction" problem

common in centralized exchanges.

2. Native Utility Burn

Pay and destroy

For scenarios where $PACT payment protocol service fees or value-added service fees are directly used, we will implement a 100% total destruction policy.

- Service fee destruction: When a user pays a $PACT guarantee fee to enjoy a rate discount, the $PACT is destroyed immediately and does not enter the treasury circulation.
- Ecoticket destruction:
    - Advertising costs for business ranking.
    - Advanced call fee for third-party API.
    - IoT device registration fee.
    - *All of the above costs are paid by $PACT and destroyed directly.*

3. Slashing Burn

The Cost of the Bad Actor Becomes the Dividend of the Owner

In the DAN arbitration network, to ensure fairness, we convert the assets of malicious actors into fully networked cryptocurrency fuel.

- Arbitrator misconduct: If an arbitrator is found to have accepted bribes or cast votes with malicious intent, their $PACT stake will be slashed. Half of the stake will be injected into the insurance fund, while the other half will be directly destroyed.
- False Claims: If a buyer is found to have maliciously initiated a false dispute to extort the seller, the arbitration deposit they pledged will be forfeited and destroyed.

4. Dust Cleanup

To optimize ledger size and reduce invalid UTXO accounts, the protocol enables users to convert small amounts of $PACT (Dust)—insufficient to cover Gas fees—into Credits with a single click. These Dust tokens will be centrally aggregated and ultimately burned.

5.6 Economic Model Sustainability Analysis

In developing OmniPact economic model, we conducted an in-depth analysis of success and failure cases from both the DeFi 1.0 and 2.0 eras. Rejecting the' Ponzi scheme 'that relies on hyperinflation to sustain false prosperity, OmniPact instead establishes a sustainable closed-loop system based on' Real Yield, 'ensuring the protocol's resilience through market cycles.

1. External Revenue Sources

— Refuse to put your left foot on your right foot

Many protocols collapse because their profits are entirely dependent on token inflation (i.e., 'printing money'). When the token price drops, the APY plummets, triggering a self-reinforcing cycle of capital flight.

OmniPact core strength lies in its revenue externalities:

- Actual commercial value: The service fees are generated from specific commercial guarantee activities, including over-the-counter (OTC) transactions, e-commerce fulfillment, and outsourced payments. These transactions are primarily settled in hard currencies such as USDT, USDC, and ETH.
- Currency-independent: Regardless of $PACT price fluctuations, the protocol consistently

captures USDT/ETH revenue whenever OmniPact is used for margin trading. This hard-currency income forms a solid foundation (Bottom Support) for $PACT value.

2. Nash Equilibrium of Supply & Demand

We have established a self-regulating dynamic equilibrium mechanism to prevent the system from overheating or undercooling.

- **Scenario A: When token prices rise**
  - The value of arbitrator collateral increases, the cost of malicious activities rises, and network security improves.
  - -> Attract more high net worth transactions.
  - The rise in protocol revenue and intensified buyback and burn activities have further driven up the cryptocurrency's price.
- **Scenario B: When the token price drops**
  - Lowering the threshold for arbitrators (in fiat currency) to attract more ordinary users to participate in governance.
  - The discounted portion of the protocol service fee (calculated in $PACT) has become more affordable, encouraging more merchants to use $PACT for payments.
  - As long as commercial demand (guarantees) persists, the agreement will continue to generate USDT revenue. In this scenario, the buyback engine can repurchase and burn more $PACT at lower prices, accelerating deflation and stabilizing the token's value by reducing supply.

3. Anti-Fragility and Counter-Cyclicality

OmniPact business inherently exhibits anti-cyclical traits, outperforming even during bear markets.

- The bull market engine: Active asset trading and frequent NFT/Token swaps drive up transaction fee income.
- Bear Market Engine: As market volatility intensifies and trust collapses (as seen in the FTX incident), demand for 'decentralized collateral' and 'funds security' surges exponentially. OTC trading typically becomes more active during bear markets, ensuring OmniPact maintains stable cash flow even in downturns.

4. Avoiding the Long-Term Lock-in of the "Tragedy of the Commons"

Through the vePACT model, we integrate governance rights, dividend rights, and lock-up periods.

- Consistency of interests: Only long-term lock-up holders can obtain the highest Real Yield dividends. This process filters out short-term speculators (Mercenary Capital) and retains long-term investors genuinely committed to the protocol's development.
- Preventing sell pressure: A large portion of circulating tokens are locked in staking contracts at all times, significantly reducing potential sell pressure and volatility.

6. Business Model and Market Growth

The core business of OmniPact is 'Monetization of Trust.' Rather than relying on unsustainable token inflation to sustain operations, we capture external commercial value

(Real Yield) by providing authentic collateral settlement services. Our goal is to build a high-margin, scalable, and network-effect-driven decentralized business empire.

6.1 Detailed Business Model Analysis

6.1.1 Four Core Revenue Streams

We have developed a multi-tiered revenue engine, covering both high-frequency trading for retail investors and subscription-based services for institutional clients.

1. Base Protocol Fees-High-frequency transaction revenue

This is the cash flow cornerstone of the agreement.

- Take Rate: A fee of 0.5% to 1% is charged for each successful secured transaction.
- Suitable for: OTC token swaps, freelance milestone payments, and e-commerce purchases.
- Advantages: Compared to traditional platforms (Upwork 20%, eBay 13%), our low rates create a strong substitution effect.

2. B2B API Subscription Fee (Enterprise Licensing) -Stable Recurring Revenue (ARR)

For third-party platforms integrated with OmniPact (e.g., NFT markets, cross-border ERP), the service is billed via SaaS.

- Subscription plan:
    - Start-up Plan: Free integration with revenue sharing based on transaction volume.
    - Enterprise Plan: Monthly subscription fee of $500- $2,000, lower transaction fees, exclusive DAO customer support, and higher API call limits.
- Revenue model: As the GMV of B-end ecosystem partners grows, the agreement generates stable passive income.

3. Arbitration & Judiciary Tax-Revenue from Dispute Resolution

- Arbitration initiation fee: A 3%-5% fee of the transaction value is required to start a dispute. While most of this fee goes to the arbitrator, the platform retains a 10%-20% Administrative Cut to maintain the arbitration network.
- Appeal fees: The cost doubles with each appeal, while the agreed commission rate remains unchanged.

4. Value-added Services Matrix (VAS) -High-margin Long-tail Revenue

- Privacy Protection Fee: An additional 0.1% privacy fee applies when using Privacy Matrix (ZK/TEE) for encrypted transactions.
- Urgent arbitration: Pay additional fees to receive high-priority processing on the DAN network.
- IoT device registration fee: Each RWA hardware device connected to the network must pay a one-time registration fee (e.g., $10 per device).

6.1.2 Cost Structure Analysis (Cost Structure)

As a decentralized protocol, OmniPact features an exceptionally cost-effective asset-light structure.

- Fixed costs (Variable Costs):
    - Prophecy call fee: Gas fee paid to Chainlink/Omni-Link nodes, which increases

linearly with transaction volume.
    - Storage fee: A fee for permanent storage with Arweave.
    - Optimization strategy: By deploying Layer 2 and implementing batch processing, marginal costs are reduced to near zero.
- Fixed Costs:
    - Security audit and bounty: ongoing contract audit fees (approximately $200k-$500k per year).
    - Core Development and Marketing: Compensation packages for DAO core contributors and marketing budget allocation.
- Profit Margin: With no need for a large customer service team or centralized server maintenance costs, OmniPact's gross margin is projected to remain above 80%.

6.1.3 Profitability Stages Planning

We don't aim for instant profitability upon launch, but instead follow the classic J-shaped growth curve of internet platforms.

Phase	objective (Goal)	policy (Strategy)	Financials
P1: User subsidy period (Year 1)	**Gaining market share**	Implementing "zero fees" or "transaction mining"; providing substantial subsidies for Gas fees.	**net deficiency** (Burn Rate for customer acquisition)
P2: The Period of Ecological Balance (Year 2)	**balance between income and expenditure**	Stop subsidies; charge for advanced features; B-end SaaS revenue starts to scale.	**break even** (Breakeven)
P3: The Profit Harvesting Period (Year 3+)	**maximization of value**	Rate normalization; network effects creating barriers; RWA implementation on a large scale.	**high profit** (Repurchase agreement activation)

6.1.4 Single-user ARPU Uplift Path

How can we maximize revenue from a single user? We developed the 'Land & Expand' strategy.

1. Entry: Users enter the ecosystem through free or low-frequency OTC transactions (e.g., U exchanges).
    - *ARPU: $0.5*
2. Retention: Users find OmniPact user-friendly and start using it to handle freelance mid-frequency orders or cross-border payments.
    - ARPU: $50 (cumulative)
3. Upselling: Users pay for credit display services or privacy encryption features to boost their order conversion rate.

    - *ARPU: $100*
4. B2B conversion (Enterprise): Users integrate OmniPact into their e-commerce platforms or DAOs, becoming B2B subscription clients.
    - ARPU: $5,000+ (annual fee)

6.1.5 Market Size & Growth Forecast

OmniPact is positioned at the intersection of three trillion-dollar markets.

- **TAM (Total Addressable Market) - $20 Trillion+**
    - Global cross-border trade and B2B payment market.
    - Vision: OmniPact to become a SWIFT-like underlying settlement protocol.
- **SAM (Serviceable Market) - $450 Billion**
    - The global gig economy: $350 billion.
    - Small and medium-sized cross-border e-commerce sellers: $100 billion.
    - Objective: To capture a portion of the market share currently held by Upwork, Fiverr, and PayPal.
- **SOM (Available Market) - $50 Billion (Web3 Native)**
    - The crypto over-the-counter (OTC) market sees daily trading volumes exceeding $10 billion.
    - Web3 service outsourcing: DAO-based compensation and code audit fees.
    - Strategy: This is our primary battleground, with a three-year target to capture 20% market share in this vertical sector.

Growth Projection:

Based on conservative estimates, we assume the Year 3 agreement captures 1% of the SOM market (equivalent to $500 million in GMV):

- Annual Transaction Volume (GMV): $500,000,000
- Agreement Revenue (1%): $5,000,000
- Burn @ 40%: $2,000,000/year

This demonstrates OmniPact exceptional commercial potential.

6.2 Market Expansion Strategy

OmniPact market entry strategy follows a dual-helix approach: transitioning from Web3-native to Web2-integrated ecosystems, and shifting from consumer-facing viral spread to business-facing ecosystem embedding. We divide market expansion into three strategic phases, designed to maximize network effects with minimal customer acquisition costs (CAC).

Phase 1: Initial Ice-Breaking-Aggressive On-Chain Incentives (The Ignition: Aggressive On-Chain Incentives)

Objective: To acquire over 10 million active seed users within 6 months and establish preliminary liquidity and credit data.

1. "Omni-Drop" Interactive Airdrop Program: We avoid meaningless cash giveaways. All incentives are designed to cultivate user habits.

- Escrow Mining: During the first 6 months after the launch, users who complete each secured transaction will receive SBT credit points and can redeem SBI credit points for PACT token airdrops when TGE.
- Vampire Attack: A targeted airdrop targeting active addresses of competitors, such as

certain centralized OTC platforms or users who have suffered from Rug Pull attacks.

- Viral Referral Loop:
    - A two-way reward system: Inviters earn 10%-20% of the invitee's transaction fees for the next 3 years as commission (directly deposited into the wallet). This incentivizes KOLs and community leaders to promote voluntarily.

2. "Safe Harbor" Marketing Campaign

- In collaboration with leading wallets (e.g., MetaMask, Trust Wallet) and security firms, we released the *Web3 Fraud Prevention White Paper*. By educating users on *how to guard against OTC scams*, we strategically positioned OmniPact as the sole recommended security solution.

Phase 2: Mid-term Penetration — B2B2C Ecosystem Embedding (The Expansion: Ecosystem Embedding)

Objective: By integrating the SDK, OmniPact will become the go-to component for Web3 traffic entry points.

1. "Powered by OmniPact Alliance Program We understand that users prefer to complete transactions on their existing platforms rather than download new apps.

- Wallet integration: Develop a MetaMask Snaps plugin or build it into mainstream multi-chain wallets. Users can activate OmniPact directly by checking 'Enable secured mode' on the transfer page, without redirection.
- Vertical platform collaboration:
    - NFT Market: Provides anti-counterfeiting safeguards for NFT trading platforms.
    - DAO tools: Integrated with DAO payment platforms (e.g., Utopia, Parcel) as the underlying escrow layer.
- B-side Incentive: Developers/platforms integrating OmniPact SDK will receive 20% of transaction fees as commission. This creates strong incentives for B-side partners to promote our services.

2. Developer Grants Program

- A $5 million ecosystem fund will be established to support developers in building innovative applications based on the OmniPact protocol, including credit-based collateral-free lending DApp and decentralized home service platforms.

Phase 3: Long-term Strategy — Global Expansion and RWA Implementation (The Ubiquity: Globalization & RWA)

Objective: To break through the barriers between dimensions and enter the trillion-dollar real economy market.

1. Global Ambassadors and Localized Guilds

- Omni-Guilds: Establish localized guilds in regions with active cross-border trade.
    - Key initiatives: Recruiting local arbitrators to resolve language barriers and training local merchants in USDT cross-border payments.

2. Industry Standard Development and Compliance (Standardization & Compliance)

- Collaborate with the International Chamber of Commerce or logistics giants like DHL/Maersk to develop a blockchain-based bill of lading standard.
- To apply for a regulatory sandbox license in the crypto-friendly zone, we are exploring

the integration of OmniPact on-chain arbitration results with local court enforcement systems, thereby creating a legal closed loop of' on-chain adjudication, off-chain execution'.

6.3 User Growth and Retention

OmniPact growth strategy avoids mindless traffic accumulation, instead prioritizing the acquisition of seed users who are high-value, highly engaged, and highly credible. By building a growth hacking funnel and an education empowerment system, we transform traffic into loyal co-builders of the ecosystem.

6.3.1 Early Seed User Acquisition: The Cold Start Campaign

Objective: To recruit 10,000 core traders and 500 founding arbitrators within the first 3 months.

1. Vertical Niche Targeting: Instead of searching for a needle in a haystack, we focus on penetrating deep into specialized vertical markets. Our strategy zeroes in on three high-risk, high-demand scenarios:

- OTC trading community: Engage with major OTC margin groups on Telegram/Discord. Offer OmniPact Zero Fees Experience Cards' to group admins and active traders, replacing manual margin calls with automated smart contracts to alleviate concerns about administrators disappearing.
- Web3 Freelancers: Targeting developers and designers on Bounties and DeWork. By offering hackathon prize sponsorships and mandating OmniPact for settlements, they can experience the product's seamless operation while claiming their rewards.
- Game item dealer: Specializes in high-value NFT and GameFi equipment trading, offering a 'recovery-proof' guarantee template to attract professional merchants.

2. "Trust Migration" Program

- Reputation Mapping: Enables users to link their Web2 accounts (e.g., Upwork, eBay, Github) via oracles. If they hold high reputation on Web2 platforms (e.g., Upwork Top Rated), OmniPact will directly grant them an initial Lv3 reputation tier.
- Strategic significance: This drastically reduces the migration costs for premium merchants, enabling them to enter the Web3 ecosystem with their established reputation intact.

3. Ambassador and Recommendation Fission (Omni-Ambassadors)

- Launch the Global Trust Ambassador program. Recruit 50 influential KOLs or community leaders in each region.
- Dual Incentives: For each verified user who completes their first order, the Ambassador will receive both a $PACT reward and a permanent 10% commission on the user's future transaction fees.

6.3.2 User Retention Mechanism: The Reputation Moat

In Web3, Token incentives often attract users who are quick to join but just as fast to leave. OmniPact leverages the SBT reputation system to create exceptionally high switching costs.

1. Sunk Cost & Assetized Reputation

- Every Omni-Score earned by users on OmniPact is backed by real-money transactions.

- The logic is simple: When a merchant holds the 'Gold Store' certification and enjoys ultra-low fee discounts, they're unlikely to abandon OmniPact and rebuild trust from scratch on a new platform. This SBT data forms the most formidable moat.

2. Gamification-based Growth System

- Introduce the "Badges" system. Examples include: "100 Orders Complete" (fulfilling 100 orders), "Zero Dispute Master" (no disputes in 50 consecutive orders), and "Divine Judge" (98% arbitration win rate).
- These badges are not just a badge of honor—they also grant real in-game weight and benefits, motivating players to level up their accounts as if they were in an RPG.

6.3.3 User Education and Empowerment Program

The high barrier to Web3 engagement is the primary cause of user attrition. Through a comprehensive education system, we transform complete novices in the crypto industry into certified experts.

1. Omni-Academy (Trust Academy) establishes official learning centers offering courses from beginner to advanced levels:

- Newbie Zone: 3-minute video tutorials — "How to Securely Purchase a Service with USDT" and "How to Initiate a Fair Arbitration".
- Advanced Zone: Arbitrator Training Course. Covers the basics of evidence science, blockchain forensics tools, and game theory voting strategies.

2. Mock Court

- Run the "Mock Trial" DApp on the Testnet. Users can experience the full dispute resolution process by acting as plaintiff, defendant, and arbitrator without financial risk.
- Certification for the role: Only users who pass the exam in the mock trial can obtain the qualification certification as a mainnet arbitrator.

3. Developer Empowerment

- Provide "OmniPact Cookbook": containing 50+ plug-and-play code recipes.
- Regularly host "Trust-Build" hackathons to reward developers who create innovative guarantee solutions, such as leasing, lending, and insurance.

6.4 Community Operations & Governance

OmniPact ultimate goal is to achieve full decentralized autonomous governance (DAO). We believe OmniPact can only become a truly neutral global infrastructure by transferring ownership and governance rights to the community. To this end, we have designed a governance framework featuring' decentralized checks and balances, proposal-driven mechanisms, and contribution-oriented principles.'

6.4.1 DAO Organizational Structure: Separation of Powers (DAO Organizational Structure)

To strike a balance between decision-making efficiency and decentralization, OmniPact DAO employs a hybrid framework combining a Council, a General Assembly, and specialized committees.

1. The General Assembly-Legislative Power

- Members: All $vePACT (staked tokens) holders.
- Power: The highest decision-making authority in the protocol. It can initiate and vote on OIP proposals, determine protocol parameters (such as fees and burn ratios), and allocate treasury funds.
- Mechanism: 1 Token = 1 Vote (with a quadratic voting cap to prevent whales).

2. Guardian Council-The veto power and safety valve

- Members: Composed of 9-15 highly credible entities elected by the community (including core development teams, top arbitrators, security audit bodies, and early investors).
- Veto Power: The right to veto, but not to propose.
- Responsibilities: When detecting "malicious governance attacks" (e.g., hackers attempting to steal treasury funds through proposals) or critical code vulnerabilities, the council may suspend protocols or veto proposals, functioning as a security fuse.

3. Professional Committees (Sub-DAOs) -Executive Authority

- The Dev Guild oversees core code maintenance and Grant reviews.
- The Arbitration Commission (Juror Guild) is responsible for establishing ethical standards for arbitrators and handling complaints against them.
- Growth Guild (Marketing Committee): Manages the marketing budget and recruits global ambassadors.

6.4.2 Proposal & Voting Process: OIP Standard

Any major changes to the protocol must follow the OIP (OmniPact Improvement Proposal) standard procedure, a rigorous on-chain legislative process.

Step 1: Ideation & Discussion

- Community members initiate discussions on the Discourse forum. If they gain enough attention, the discussion can be upgraded to a draft.

Step 2: Official Proposal (OIP Submission)

- The proposer must hold a certain amount of $vePACT (e.g., 100k+) as the minimum threshold. The proposal must include detailed technical specifications, code changes, and an economic impact analysis.

Step 3: Snapshot Signaling

- Conduct a Gas-free pre-vote on the Snapshot platform to quickly pass community intent testing and filter out low-quality proposals.

Step 4: Chain the final vote (On-Chain Ratification)

- When Snapshot's approval rate exceeds 60%, the proposal is processed through on-chain governance mechanisms such as Governor Bravo.
- All $vePACT holders will vote within 3-7 days.

Step 5: Timelock & Execution

- If the proposal receives a majority vote and meets the quorum, it will enter a 48-hour timelock period.
- After the public notice period ends, the smart contract automatically executes code changes (such as modifying rate parameters) without any human intervention.

6.4.3 Community Incentives & Contributor Program

People are at the heart of DAO. To attract top global talent to build OmniPact, we launched the 'The Pact Builders' contributor program.

1. Global Ambassadors Program

- Responsibilities: Manage community operations locally, translate white papers, and organize offline Meetups.
- Incentives: Monthly rewards of $USDT + $PACT based on KPI (active users, event frequency). Top performers may receive exclusive NFT badges.

2. Bounty Hunters

- Role: Fix bugs, develop third-party plugins, and optimize SDK documentation.
- Incentives: Post tasks on Gitcoin or Questbook for piece-rate payment. For significant contributions (e.g., critical vulnerability discovery), Immunefi offers high rewards.

3. Content Creators Incentives (Content Creators)

- Role: Create tutorial videos, write analytical articles, and design memes.
- Incentive: Launch a 'Creator Fund' with automated rewards based on content views and conversion rates.

4. Governance Participation Rewards (Governance Rewards)

- To tackle voter apathy, all $vePACT holders participating in the OIP vote will receive POAP badges and governance airdrop rewards.

6.5 Operational Milestones and KPI Projections

OmniPact operational strategy is data-driven. We have divided the next 36 months into three key growth phases, with clear quantitative goals (KPI) to ensure the team remains focused on core business growth.

Stage 1: Validation and Cold Start

Phase: Month 0 to Month 6. Objective: To validate PMF (Product-Market Fit) and build a core seed community.

- User growth:
    - The number of registered wallet addresses has exceeded 10,000,000+.
    - The platform has surpassed 6 million daily active users (DAU).
- Business data:
    - The cumulative gross market value (GMV) of the collateralized transactions reached $10,000,000.
    - Successfully handled over 500 dispute cases, with an arbitration satisfaction rate exceeding 90%.
- ecological construction：
    - Recruit 500 creation arbitrators and train them.
    - SDK integration has been completed with five specialized Web3 platforms, including GameFi Market and OTC community platforms.
    - The combined community size of Twitter and Discord has surpassed 200,000 members.

Stage 2: Growth and Network Effects

Phase: Month 7-Month 18. Objective: Achieve scaled expansion through a B2B2C strategy

and complete the token economy closed loop.

- User growth:
  - The number of active users has exceeded 50,000,000+.
  - The number of B-end merchants/integrators has exceeded 100.
- Business data:
  - The cumulative GMV has exceeded $100,000,000.
  - The agreement's monthly revenue exceeded $2,000,000, triggering the $PACT buyback and destruction program.
- Capital and Market:
  - After completing the TGE (Token Generation Event), log in to a leading CEX (e.g., Binance, OKX, or Coinbase).
  - Launch the Global Ambassador Program, covering 10 key country markets in Southeast Asia, Latin America, Africa and beyond.

Stage 3: Monopoly and Entity Integration (Dominance & RWA Integration)

Phase: Month 19-Month 36. Goal: Establish Web3 trust infrastructure standards and achieve large-scale deployment of RWA services.

- User growth:
  - More than 200 million users worldwide.
  - Omni-ID has become the go-to 'credit passport' for Web3 users.
- Business data:
  - The annualized GMV surpassed $10 billion.
  - RWA physical asset transactions account for over 30% of its total.
- Ecological barriers:
  - The DAN arbitration network has over 10,000 nodes, achieving fully decentralized operation.
  - DAO governance manages protocol parameter permissions to enable community autonomy.

7. Ecological Development Plan

OmniPact ultimate vision is to become the "Android of Trust" for the Web3.0 economy. By opening atomic capabilities, establishing an ecosystem fund, and building a global alliance, we are committed to creating a vast, self-evolving, and mutually beneficial business ecosystem.

7.1 Short-term Ecological Goals (1-2 years): Foundation Building and Breakthrough (Short-term Goals: Foundation & Breakout)

The first two years of OmniPact are a pivotal 'Critical Mass Building' phase. Our strategy focuses not on indiscriminate industry-wide coverage, but on establishing absolute dominance in specific high-pain verticals and building an initial flywheel effect connecting developers, merchants, and users.

We divide this phase into two core campaigns:

Year 1: Genesis Validation & Developer Cold Start

In the first year, our goal is to prove OmniPact as the most user-friendly staking tool in Web3 and attract the first batch of 'first movers' developers.

- **Technical infrastructure goals:**
  - Completed protocol deployment on three major blockchains: Ethereum, BNB Chain, and Arbitrum.
  - The release of OmniPact SDK V1.0 ensures developers can complete integration within 4 hours.
  - The base version of Omni-Link Prophet is now available, seamlessly integrating data interfaces between GitHub (code delivery) and FedEx/DHL (logistics delivery).
- **Developer and app ecosystem:**
  - Host two global hackathons to incubate at least 10 OmniPact-native DApps, including decentralized service marketplaces and P2P lending demos.
  - Recruit over 50 core contributors to join Developer DAO and build a vibrant tech community.
- **Vertical feature breakthrough:**
  - It commands 5% market share in both Web3 freelance services and OTC token trading, establishing itself as the preferred collateral solution in this sector.

Year 2: Cross-Chain Fusion and Enterprise Penetration

In the second year, we will leverage Token Economy Incentives (TGE) to accelerate ecosystem expansion, break through blockchain boundaries, and begin penetrating the Web2 traffic ecosystem.

- **Connectivity goal:**
  - The launch of the X-Escrow cross-chain module bridges the gap between non-EVM ecosystems like Solana and Aptos, enabling seamless cross-chain fund transfers.
  - We have established deep partnerships with three leading wallets (including MetaMask, OKX Wallet, and Trust Wallet), integrating the OmniPact Secure Transfer feature into their platforms to directly serve millions of users.
- **B-end commercialization:**
  - We have partnered with over 20 mid-sized B2B entities (including GameFi guilds, NFT exchanges, and independent e-commerce platforms), generating steady revenue through our API subscription model.
  - The first RWA benchmark case demonstrates IoT-Anchor closed-loop capability.
- **Governance evolution:**
  - Launch the public beta of the DAN arbitration network, featuring at least 1,000 active arbitrator nodes to enable decentralized dispute resolution.

Short-term Ecosystem KPI Forecast Matrix (Year 1-2)

To quantify ecological prosperity, we established the following core indicators:

Key Metrics	Year 1 Goal (Foundation)	Year 2 Objective (Expansion)

Key Metrics	Year 1 Goal (Foundation)	Year 2 Objective (Expansion)
Number of integrated DApps/platforms	10+	50+
Active developers	200+	1,000+
Supported asset types	ERC-20, ERC-721	Full-chain assets + basic RWA
arbitration node distribution	10 countries/regions	35 countries/regions
amount of funds disbursed by the ecological fund	$2,000,000	$10,000,000

7.2 Medium-term Ecological Layout (3-5 years): Fusion & Standardization

After completing the initial technical groundwork and vertical scenario validation, OmniPact will shift its strategic focus to 'Mass Adoption' and 'Integration with the Real Economy' in the mid-term. Our goal is to establish OmniPact as the de facto standard for global decentralized commercial settlement within 3-5 years.

We will achieve this ambitious blueprint through the following four strategic pillars:

1. RWA and Supply Chain Finance: A Deep Dive

Strategic objective: To integrate blockchain trust into the trillion-dollar global trade market.

From Year 3 to 5, OmniPact will expand beyond digital assets to fully handle complex real-world transactions.

- Digital Bill of Lading (DBL):
  - Partnering with global logistics leaders and port operators to promote OES-standard e-Bills of Lading. By linking cargo ownership to OmniPact NFT, this enables instant 'cash-on-delivery' settlements, replacing traditional inefficient Letters of Credit.
- IoT Finance (IoT-Fi):
  - Deploy millions of IoT-Anchor-enabled smart devices. From shared rental equipment (e.g., EV charging stations) to cold chain transport containers, these devices report status directly via the OmniPact protocol and trigger fund transfers, building a Machine Economy.

2. "OmniPact Inside" Enterprise Integration

Strategic objective: Enable seamless integration of Web2 enterprises into Web3 financial networks.

We aim to shatter the stereotype that 'Web3 is just for tech enthusiasts' by leveraging middleware technology to bridge traditional business software.

- ERP system connector:
  - Develop official plugins for mainstream ERP systems including SAP, Oracle, and Salesforce.
  - Scenario: A multinational manufacturing company using SAP can directly issue a USDC-backed secured order to suppliers through its procurement system, with OmniPact handling all backend blockchain interactions, currency hedging, and compliance audits.
- fiat/coin hybrid gateway:
  - Establish a hybrid payment channel with compliant service providers like Stripe

and MoonPay. When buyers pay in fiat currency (USD/EUR), OmniPact automatically converts it into stablecoins in the backend, ensuring sellers ultimately receive either fiat or cryptocurrency—achieving seamless two-way transactions.

3. AI-Driven Arbitration Scaling

Strategic objective: To leverage AI to overcome efficiency bottlenecks in decentralized justice.

With tens of millions of users, manual arbitration alone cannot handle the sheer volume of minor disputes. We will introduce AI-Juror (AI-arbitrators).

- LLM-assisted adjudication: Leveraging fine-tuned large language models (LLMs)for case preprocessing. AI analyzes thousands of pages of chat logs and code repositories to generate 'evidence summaries' and 'judgment recommendations,' with human arbitrators performing only final review.
- AI-powered automated small claims court: For disputes under $100, both parties may fully authorize AI arbitrators to make instant rulings, dramatically cutting legal costs.

4. Developing industry standards (Standardization)

Strategic objective: To establish OES as the ISO standard for trust-based transactions in Web3.

- OES Protocol Standardization: Facilitate the inclusion of OES (Omni Escrow Standard) into the EIP (Ethereum Improvement Proposals) standardization process, establishing it as an industry-wide escrow interface standard.
- Establish a 'Trust Alliance': Collaborate with leading DeFi protocols, exchanges, and auditing firms to form a Web3 credit consortium. This alliance will jointly maintain the Omni-ID blacklist database, ensuring that 'dishonest actors' face severe restrictions across the entire Web3 ecosystem.

Mid-term Ecosystem KPI Forecast (Year 3-5)

Core Metrics	Year3 target	Year5 target
annual gross merchandise volume	$ 1 billion （Billion）	$ 10 billion +
RWA business proportion	15%	40%
Number of B2B enterprises integrated	500+	5,000+
active arbitration node	5,000	20,000
repurchase amount	$5,000,000 per year	$30,000,000 per year

7.3 Long-term Ecological Vision: The Internet of Value

Looking ahead, OmniPact ambition extends beyond building a successful Web3 unicorn company—it aspires to become the' Trust TCP/IP Protocol' of human digital civilization. Over the next five years and beyond, we are committed to creating a frictionless, borderless, and intermediary-free global value network.

1. Trust as a Public Utility: The Protocol for Public Water and Electricity

We foresee that in the future, 'trust' will no longer be a private commodity monopolized by large corporations (such as banks and e-commerce giants), but rather a public infrastructure as affordable and readily accessible as electricity and the internet.

- Zero marginal cost: OmniPact reduces the cost of establishing business trust between two

completely unfamiliar entities (human or machine) to near zero.

- Global collaboration equality: Whether you're on Wall Street in New York or in rural Africa, Omni-ID empowers you to participate equally in the global economic division of labor. OmniPact will eliminate commercial discrimination and unleash the suppressed productivity of billions.

2. The Cornerstone of the Machine Economy (Infrastructure for the Machine Economy)

Over the next decade, AI agents and robots will dominate economic activities. However, AI operates without laws or ethics, relying solely on code.

- M2M (Machine-to-Machine) settlement layer: OmniPact will serve as the standard protocol for AI agents to collaborate, conduct transactions, and provide services.
    - Scenario: An autonomous vehicle (lessor) and an AI-powered logistics dispatching algorithm (lessee) automatically execute a transportation contract, secure funds via OmniPact, and settle automatically upon task completion—without any human intervention.
- Self-governing economies: IoT devices will have their own wallets and digital identities (DID), enabling 'self-sustaining and self-generating revenue.' OmniPact serves as the legal framework sustaining these systems.

3. The Passport of Network States

As physical borders blur, Network States founded on shared values will emerge. OmniPact System of Basic Trust (SBT) will evolve into the new digital passport.

- Reputation is money: A person's credit score will be more valuable than their bank deposits. A high Omni-Score will become the hard currency for entering elite communities, obtaining unsecured loans, and even participating in global governance.
- Sovereign individuals: Users will regain full control over their data and assets, free from the restrictions or censorship imposed by any single platform.

4. Hyperstructure & Immortal DAO

Ultimately, OmniPact will transition from relying on its founding team and even the DAO human governance to evolving into a' Hyperstructure.

- Self-evolution: The protocol will evolve into a self-sustaining digital ecosystem, much like the Bitcoin network. It operates autonomously through mathematical rules and self-reinforces via economic incentives.
- Public goods: They belong to all mankind and serve all mankind. No one can shut them down, and no one can monopolize them.

7.4 Developer Ecosystem Support Plan

At OmniPact, we believe developers are the core value creators. Our 'Developer-First' strategy is dedicated to building the most developer-friendly and profitable ecosystem in the Web3 industry.

We launched the OmniPact Builders Program, which identifies and nurtures the next unicorn application through three key dimensions: funding, technology empowerment, and market support.

1. Financial Engine: The $50M Ecosystem Fund

We pledge to allocate 15% of the total token supply and 20% of the protocol treasury revenue to the ecosystem fund, providing "full lifecycle" funding support for developers at various stages.

- Micro-Grants:
    - Amount: $2,000- $10,000.
    - Target: Individual developers, hackathon teams.
    - Purpose: For developing MVP (Minimum Viable Products) or open-source tools (such as SDK language packs). No equity returns required, only open-source code.
- Strategic Incubation:
    - Amount: $50,000- $200,000.
    - Target: Startups with a complete team and business plan (e.g., decentralized outsourcing platforms in vertical sectors).
    - Support: In addition to funding, we provide office space, legal consultation, and guidance on designing token economy models.
- Gas Rebate Program:
    - For high-frequency trading DApp, the protocol will subsidize 50%-100% of Gas fees generated by users, lowering the barrier to entry for new customers.

2. Technology Empowerment: Zero-Friction Development Experience (Zero-Friction DX)

Our goal is to enable Web2 developers to integrate OmniPact within 4 hours.

- Omni-Kit:
    - We provide ready-to-use UI components (front-end component library), enabling developers to simply copy and paste the code to embed features like a 'guaranteed payment button' or 'arbitration progress bar' on their web pages.
- Debug Studio:
    - Provide a cloud debugging platform based on the Fork mainnet environment. Developers can simulate complex edge scenarios such as 'buyer refusal' and 'arbitration intervention' with one click to quickly troubleshoot bugs.
- Audit Fast-Track:
    - Projects developed using OES standard templates qualify for a 50% discount on audit fees and priority scheduling from OmniPact partner security firms such as CertiK/SlowMist.

3. Growth System: Hackathon and Certification (Growth & Certification)

- Trust-Build Global Hackathon:
    - The event is held quarterly in Denver, Singapore, Berlin, and other cities, featuring three specialized tracks: RWA, DeFi, and DAO tools.
    - Rewards: In addition to cash prizes, winning projects will be directly invited for investment interviews by Binance Labs or VC partners.
- Certified OmniPact Developer Program:
    - Launch official technical certification. Developers who pass the assessment will receive an SBT badge.
    - Benefits: Listed on the official 'Recommended Service Providers List', we are prioritized for B2B clients' customized development needs (e.g., ERP integration).

4. Traffic and Marketing Support

- Launchpad IPO: Premium ecosystem projects can conduct Initial Coin Offerings (ICO) or Initial Exchange Offerings (IEO) through OmniPact official community, directly

engaging over 500,000 active trading users.

- Joint Promotion: Official Twitter, Discord, and a media network will provide free exposure support for the ecosystem project.

7.5 Ecosystem Partner Recruitment and Cooperation Models

OmniPact is committed to building a global trust alliance that is open and mutually beneficial. We warmly invite leading enterprises from both Web3 and Web2 sectors to join us in reshaping the fundamental principles of business transactions. To accommodate diverse partners, we have designed flexible and high-reward collaboration models.

1. Target Partners and Profile

We focus on recruiting three key categories of core partners:

Category A: Platform Integrators — Traffic Entry Points

- Target: NFT exchanges, decentralized over-the-counter (OTC) platforms, Web3 freelancer marketplaces, GameFi guilds, and independent e-commerce platforms (Shopify/WooCommerce).
- Value: You have users and transaction scenarios, but lack secure fund custody and arbitration capabilities.
- Key feature: The OmniPact SDK is integrated, with 'guaranteed payment' serving as the platform's foundational infrastructure.

Category B: Professional Service Nodes — Trust Anchors

- Target: International law firms, blockchain security audit firms, KYC/AML compliance service providers, and intellectual property agencies.
- Value: You have professional legal and forensic expertise.
- Collaboration Opportunity: Become an 'Expert Arbitration Node' or 'Verification Node' in the DAN network to handle high-value dispute cases.

Category C: RWA Infrastructure Partners-the bridge to reality

- Target: International logistics giants (DHL/FedEx/Maersk), IoT hardware manufacturers (smart locks/sensors), and payment gateways (MoonPay/Stripe).
- Value: You control the data and capital flows in the physical world.
- Integration options: Access the API to Omni-Link predictor, or pre-install IoT-Anchor firmware to enable IoT asset blockchain integration.

2. Cooperation Models & Benefits

We have abandoned the traditional "buying and selling relationship" and established a deep interest sharing mechanism instead.

Model 1: Revenue Sharing Model

— For Class A platform integrators, this is the most direct economic incentive.

- Mechanism: For all OmniPact guaranteed transactions initiated through the partner platform, the service fee (0.5%-1%) charged under the agreement will be proportionally refunded to the partner.
- Profit sharing ratio:
    - Standard Partner: 20% commission.

    - Strategic Partner: 30% -50% commission (requires $PACT deposit).
- Win-win Logic: Partners gain zero-cost monetization channels while boosting user retention (as funds are more secure).

Mode 2: Node Incentive Model

— Applicable to Class B professional service providers

- Mechanism: Partners pledge $PACT to become advanced arbitration nodes.
- income：
    - Arbitration fees: Exclusive to high-value cases (e.g., commercial disputes exceeding $100,000).
    - Consultation fee: We provide value-added services such as "customized compliance guarantee templates" for enterprise users and charge a consultation fee.

Model 3: Tech & Branding

— Applicable to C-class infrastructure partners

- Mechanism: Jointly develop industry standards (e.g., "Blockchain Electronic Bill of Lading Standards").
- rights and interests：
    - Brand Endorsement: Featured as an 'Official Trusted Partner' on OmniPact official website and in its white paper.
    - Technology interoperability: OmniPact will prioritize acquiring or integrating data services from partners, thereby attracting additional Web3 customers.

3. Onboarding Process

To enhance collaboration efficiency, we have launched the OmniPact Partner Portal.

1. Apply: Submit company details and a letter of intent.
2. Review: The OmniPact Ecosystem Committee completes the qualification assessment within 48 hours.
3. Integration (Integrate): Get a dedicated API Key and personalized one-on-one guidance from the technical support team.
4. Launch: Joint marketing campaign to officially roll out the service and activate the real-time revenue-sharing dashboard.

8. Risk Management and Legal Compliance

As crypto assets are increasingly integrated into mainstream global financial regulatory frameworks, OmniPact recognizes that only protocols built on a solid compliance foundation can achieve long-term sustainability. This chapter presents our in-depth analysis of the global regulatory landscape and demonstrates how OmniPact leverages technological and architectural innovations to meet regulatory requirements while preserving its decentralized ethos.

8.1 Analysis of Global Regulatory Compliance Status

The current global Web3 regulatory landscape is characterized by fragmentation, volatility, and tightening. Significant legal distinctions between jurisdictions regarding DeFi (decentralized finance) and DAO pose complex challenges for global protocols like OmniPact.

1. Analysis of Three Major Regulatory Geographical Sectors (Geo-Regulatory Landscape)

We divide the global market into three strategic regions based on the clarity and friendliness of regulatory policies:

Area A: Clearly defined as the Regulated & Friendly zone

- Representative regions: European Union (MiCA framework), Hong Kong (VASP system), Singapore (MAS), United Arab Emirates (VARA).
- Current status: These regions have established clear legal frameworks for digital assets, such as the EU Markets in Crypto-Assets Regulation. They recognize the legitimacy of crypto assets but impose strict requirements on KYC/AML (anti-money laundering) and consumer protection.
- Impact: This is the key compliance focus area for OmniPact, where we need to apply for the necessary licenses or collaborate with licensed institutions.

B zone: Enforcement-Heavy

- Region: United States (SEC/CFTC).
- Current situation: There is no unified legislation, and the sector primarily relies on regulators " enforcement-based regulation (Regulation by Enforcement) '. The core dispute revolves around whether tokens qualify as' securities (Security)' and whether DeFi front-end platforms need to register as brokers.
- Impact: This is the most high-risk area. OmniPact must either strictly block U.S. users (Geo-blocking) or ensure the protocol architecture meets the 'non-securitization' requirement.

C Area: Grey/Restricted

- Representative regions: Chinese mainland, some Islamic countries.
- Current situation: A total ban on cryptocurrency trading or an extremely ambiguous legal status.
- Impact: OmniPact will block IP access in these regions at the technical level to circumvent legal risks.

2. Key Compliance Challenges

As a protocol for managing fund flows, OmniPact primarily faces three major categories of legal risks:

Challenge 1: Anti-Money Laundering and Counter-Terrorist Financing (AML/CFT)

- The FATF Travel Rule' mandates that service providers handling fund transfers must verify the identities of both parties involved in transactions. This inherently conflicts with the anonymity inherent to DeFi.
- Risk: If criminals exploit the protocol for money laundering, the development team could be accused of aiding criminal activities (as seen in the Tornado Cash case).

Challenge 2: Qualitative Analysis of Tokenized Securities (Securities Classification)

- Question: Under the US. Supreme Court's Howey Test, if a $PACT token is classified as a' securities,' it must be registered with the SEC; otherwise, it constitutes an illegal offering.

- Risks: The token's staking yield mechanism and governance rights may violate securities regulations.

Challenge 3: Legal Liability of Decentralized Autonomous Organizations (DAOs)

- Question: DAO are not legal entities in most countries. Who is liable for financial losses caused by code vulnerabilities? The founders, token holders, or code contributors?
- Risks: Ooki DAO precedent indicates that voting DAO members may be treated as partners and held jointly liable.

3. Regulatory Trend Forecast

We anticipate that future regulatory trends will shift from 'regulatory assets' to ** 'regulatory gateways' and 'regulatory protocol developers' **.

- The erosion of technological neutrality: Pure technical code may no longer be a safeguard, as protocol front-end operators (Operators) will face heightened compliance obligations.
- Embedded Supervision: Regulators may require smart contracts to incorporate built-in mechanisms such as 'blacklist' or 'transaction suspension' features.

Conclusion: OmniPact cannot afford an 'ostrich policy'. We must develop a hybrid architecture featuring 'technology neutrality, layered architecture, and proactive compliance' to navigate the tension between decentralized ideals and regulatory realities.

8.2 Core Compliance Principles & Strategy

OmniPact compliance philosophy is encapsulated in three pillars: "technology neutrality, proactive compliance, and tiered governance." We are committed to developing a Web3-specific compliance framework that safeguards the decentralized ethos (resistance to censorship and non-custodial) while meeting AML and compliance requirements across major global jurisdictions.

1. The Protocol-Interface Separation Strategy

We adopt best practices from the DeFi industry (e.g., the Uniswap Labs model) to strictly separate legal entities.

- The Protocol: The smart contract code deployed on the blockchain constitutes a "public infrastructure." It is open-source, immutable, and uncontrolled by any entity. Following the "Code is Speech" principle, the development team assumes no financial intermediary responsibility for the protocol's decentralized operation.
- Front-end interface (The Interface): A web or mobile application operated by OmniPact Labs or third parties. This serves as a compliant 'firewall', where the operator strictly enforces geo-blocking to block IP access from sanctioned regions (e.g., North Korea, Iran, and certain U.S. states).

2. ZK-KYC: Identity Compliance with Privacy Protection (Zero-Knowledge Compliance)

In response to the FATF Travel Rule, we have moved away from the conventional centralized KYC practice of 'uploading passport photos to servers' and instead adopted ZK-DID (Zero-Knowledge Decentralized Identity) technology.

- Off-chain authentication with on-chain verification: Users complete KYC verification locally through compliant third-party identity service providers (e.g., Sumsub, Synaps).
- Zero-knowledge proof (ZKP) on-chain: Instead of uploading users 'private data, the service provider generates a ZKP and attaches a KYC_Verified tag to the user's SBT.
- Result: OmniPact protocol only identifies that the user has passed compliance verification

and is not on the blacklist, but does not disclose the user's identity. This achieves a perfect balance between compliance review and user privacy.

3. Tiered Asset Pools: A Coexistence of Licensed and Unlicensed Assets (Permissioned & Permissionless Pools)

To serve both DeFi players and institutional clients, OmniPact has designed a dual-track asset pool:

- Public Pools:
  - Token swaps and NFT trading compatible with Crypto Native.
  - Threshold: No KYC required, only basic AML wallet address scanning (provided by Chainalysis Prophet) to block malicious addresses.
- Permissioned Pools:
  - Suitable for RWA assets, enterprise supply chain finance, and securitization tokens.
  - Entry requirement: Mandatory KYC. Only whitelisted addresses holding KYC_Verified SBT can interact with such contracts. This removes compliance barriers for traditional financial institutions (TradFi) entering the Web3 space.

4. Non-Securitization Token Design

To circumvent the SEC's securities classification risks, we strategically emphasized $PACT 'utility' while downplaying its 'investment' attributes in the design.

- Work token attributes: $PACT is emphasized as a mandatory tool for arbitration participation (License to Work), not merely a dividend certificate.
- Decentralized issuance (DOE): By implementing a Fair Launch and conducting widespread community airdrops, it prevents token concentration among a few individuals, thereby reducing the risk of failing the Howey Test.
- DAO governance: Emphasizes token voting rights, positioning them as member benefits of a 'digital cooperative' rather than corporate shares.

5. Legal Wrapper for DAO

To address the unlimited joint liability issue in DAO, OmniPact will register a non-profit foundation entity in cryptocurrency-friendly jurisdictions such as Zug, Switzerland, the Cayman Islands, or Wyoming.

- OmniPact Foundation: Acting as the DAO real-world representative, it handles legal contract signing, developer recruitment, and intellectual property management. In legal disputes, it represents the DAO as a limited liability entity to safeguard members' personal assets.

8.3 Data Security and Privacy Compliance

In building a global trust network, OmniPact strictly adheres to the "Privacy by Design" principle. We recognize that commercial contracts, logistics data, and arbitration evidence contain sensitive PII (Personal Identifiable Information). To address this, we have developed a hybrid framework compliant with the GDPR (EU), CCPA (California), and other major global data protection laws.

1. On-Chain/Off-Chain Segregation Principle

To comply with the GDPR data minimization principle, OmniPact implements a rigorous tiered data storage strategy:

- Layer 1: Stores only anonymized technical data.

    - Content: wallet address, transaction amount, timestamp, hash value of evidence files (Hash), and ZK verification result.
    - Compliance: These data are classified as 'Pseudonymized Data' and do not directly identify individuals.
- Off-chain (Omni-Vault): Stores encrypted sensitive business data.
    - Content: Contract text, chat records, logistics details, and KYC verification information.
    - Storage location: Encrypted on IPFS, Arweave, or local device.
    - Compliance: This data is under strict access control and cannot be viewed without authorization.

2. "Crypto-Shredding" as a Means to Enforce the Right to Be Forgotten

Article 17 of the GDPR grants users the right to request the deletion of their personal data. However, blockchain technology cannot delete data. OmniPact resolves this paradox by employing 'Crypto-Shredding' technology.

- Mechanism: When a user requests to exercise the 'right to be forgotten,' while the hash value on the blockchain or encrypted data blocks in the distributed network cannot be physically deleted, we can destroy the unique private key required to decrypt the data.
- Legal effect: Once the private key is destroyed, the remaining encrypted data becomes computationally unbreakable, equivalent to random gibberish. Legally, this is widely recognized as meeting the criteria for 'deletion' or 'anonymization'.

3. Zero-Knowledge Proof Realizes "Data Usable but Invisible" (ZK-Minimization)

The CCPA prohibits companies from excessive collection of user data. OmniPact employs zk-SNARK technology to achieve maximum data minimization.

- Scenario: In credit guarantee, the agreement does not require knowing the user's specific bank statements or home address.
- Implementation: Users generate Proof locally and answer only yes/no questions to the protocol (e.g., "Is the user of legal age?" "Is the user's credit score> 700?").
- Value: The agreement validates eligibility without ever accessing users' raw data, thereby avoiding legal liability for data breaches.

4. Data Sovereignty and Portability

GDPR Article 20 establishes the 'right to data portability,' a requirement that OmniPact's architecture inherently fulfills.

- User control: Users 'reputation data (SBT) and transaction history are managed by their private keys, not the platform's database. Users can package these data and authorize third-party applications (e.g., lending platforms) to use them at any time without OmniPact approval.
- Fine-grained authorization: With Privacy Matrix, users can precisely control who can access their data and for how long. For example, only arbitrators can view evidence during case proceedings, and the authorization automatically revokes after the case concludes.

8.4 Asset Trading Compliance

OmniPact is a multi-asset category universal collateral agreement. To comply with the complex global regulatory environment, we have established a tiered compliance framework based on asset

attributes (pure digital assets vs. real-world assets), ensuring every transaction operates within a legal framework.

1. Digital Assets Compliance: Anti-Money Laundering and Securities Screening (Digital Assets: AML & Securities Screening)

For crypto-native assets (Cryptocurrency & NFTs), compliance hinges on verifying the legitimacy of funding sources and defining token attributes.

- Real-time AML Blocking:
  - The protocol incorporates on-chain risk prediction mechanisms from Chainalysis and Elliptic.
  - Before creating any guarantee contract, the system automatically scans the buyer's wallet address. If the address is linked to hacking attacks, dark web markets, or sanctioned entities (OFAC SDN List), the contract will refuse to interact, blocking illegal funds at the source.
- Token Whitelisting:
  - To mitigate risks from unregistered securities issuance, OmniPact's front-end dApp is designed to support only mainstream, highly compliant assets by default, including ETH, WBTC, USDT, and USDC.
  - Long-tail tokens or newly issued DAO tokens must undergo community governance voting or submit a Legal Opinion Letter to be listed.

2. RWA Asset Compliance: The Legal Packaging (RWA: The Legal Wrapper)

The core challenge in tokenizing real-world assets (e.g., real estate, vehicles, bonds) on the blockchain is ensuring that on-chain tokens legally own their off-chain counterparts. OmniPact employs a dual anchoring solution combining Ricardian Contracts with a Special Purpose Vehicle (SPV) framework.

- Ricardian Anchoring:
  - Every RWA guarantee transaction generates an electronic legal contract (PDF), which explicitly states: "Holding this NFT grants ownership rights to the corresponding physical asset."
  - The contract's hash value is embedded in the smart contract, with its address referenced in the contract text. This legally binds the on-chain code to the off-chain legal framework, ensuring enforceability in court.
- SPV (Special Purpose Vehicle) for High-Value Assets:
  - For high-value assets such as real estate, OmniPact partners will establish a dedicated SPV (Special Purpose Vehicle) to hold the physical assets.
  - The token on the chain represents the SPV equity or debt. Even if the original asset owner goes bankrupt, the token holder's rights remain unaffected.

3. Physical Goods Compliance: Incoterms Adaptation

For cross-border e-commerce and supply chain trade, the key to compliance lies in defining the transfer of ownership.

- Digital Incoterms:
  - The OES standard template includes internationally recognized trade terms (e.g., FOB, CIF, DDP) established by the International Chamber of Commerce (ICC).
  - Scenario: When a transaction adopts FOB (Free On Board) terms, the smart contract recognizes the seller's fulfillment upon the logistics oracle's

confirmation of 'shipped' status at the departure port, transferring risk to the buyer. This ensures the contract's execution fully complies with international maritime law.

- Customs and tax compliance:
    - For B2B bulk transactions, the agreement allows uploading customs declarations and tax payment certificates as performance evidence, ensuring compliance with the tax regulations of both import and export countries.

4. Intellectual Property Compliance: IP-NFT Standards (IP Compliance)

- On-chain license issuance:
    - Based on the IP-NFT standard, creators can set detailed licensing terms (e.g., non-commercial use only, North America only).
    - This authorization clause is embedded in the NFT as metadata. If the buyer violates the terms (e.g., selling merchandise in unauthorized regions), the IP holder may initiate legal proceedings based on on-chain evidence or revoke the authorization via smart contracts.

8.5 Cross-Border Transaction Compliance Solutions

OmniPact is committed to building a borderless global trade network. Rather than challenging sovereign financial boundaries amid vastly different regulatory systems, we provide a ** "compliance adapter" ** that enables users to enjoy blockchain-powered settlement efficiency while complying with local laws.

1. Forex Controls and Capital Flow Compliance

For countries with strict foreign exchange controls (such as some emerging markets), OmniPact adopts a 'localized settlement, on-chain clearing' strategy.

- Stablecoins as a bridge:
    - The protocol's core mechanism employs US dollar stablecoins (USDT/USDC) as the global settlement currency, effectively enabling' global trade dollarization' while bypassing the inefficient SWIFT system.
- Local Compliance Gateways:
    - OmniPact does not handle fiat currency directly. We establish API connections with licensed OTC dealers and compliant exchanges (e.g., Coinbase, Binance P2P) in over 50 countries and regions worldwide.
    - Process: The buyer first converts their local currency (e.g., BRL) to USDC via a compliant gateway. The transaction then enters the OmniPact escrow system, where the seller receives the USDC. The seller subsequently converts the USDC back to their local currency (e.g., NGN) through their country's compliant gateway.
    - Compliance: As fiat-to-crypto exchanges occur through licensed local institutions, cross-border fund movements comply with the regulatory requirements of the local central bank.

2. Automated Tax Compliance Engine

Cross-border trade involves complex tax issues (e.g., VAT and Sales Tax). OmniPact now features a 'Smart Tax Allocation' function.

- Real-Time Tax Calculation:

    - Integrate third-party tax data providers (e.g., TaxBit, Koinly). When a transaction occurs, the smart contract automatically determines the applicable tax rate based on the geographical IP addresses of both parties (for example, if the buyer is in Germany, it automatically applies a 19% VAT rate).
- Automated Withholding:
    - When funds are released, the contract can be configured to automatically withhold the tax portion and deposit it into the seller's designated "tax-specific sub-wallet" or directly into the local tax authority's on-chain address (if supported).
    - Value: This significantly reduces compliance costs for cross-border e-commerce sellers, preventing account freezes caused by tax violations.

3. Global Sanctions Screening and Geo-Fencing

To comply with the sanctions requirements of OFAC (Office of Foreign Assets Control, U.S. Treasury Department) and the EU, OmniPact implements strict geographical and identity controls at its front end.

- IP geo-fencing (Geo-Fencing):
    - The frontend dApp automatically detects and blocks access requests from sanctioned regions, including North Korea, Iran, and Crimea.
- Sync the blacklist dynamically:
    - The protocol maintains a real-time on-chain blacklist synchronized with the OFAC list. Any funds associated with blacklisted addresses cannot enter OmniPact collateralized contracts. This mechanism is embedded in the smart contract's compliance layer, ensuring that even if users bypass the frontend and directly invoke the contract, the transaction will be reverted.

4. Trade Authenticity Verification

The essence of Anti-Money Laundering (AML) regulation lies in verifying the authenticity of transaction backgrounds. OmniPact employs RWA technology to deliver audit trails with greater transparency than conventional banking practices.

- On-chain Audit Trail:
    - Every fund transfer is strictly tied to specific business documents, including contract hashes, logistics tracking numbers, and acceptance reports.
    - When regulators or banks require audits, users can export a 'Proof of Business' package containing all transaction and information flows with a single click. This transforms cryptocurrency transfers from anonymous digital transactions into verifiable commercial settlements.

8.6 Compliance Iteration and Adaptation Mechanism

The global crypto regulatory landscape is undergoing dramatic upheaval. OmniPact rejects rigid 'one-size-fits-all' approaches, instead developing a programmable, modular, and dynamically evolving compliance framework. This ensures the protocol can seamlessly adapt to the latest legal requirements across jurisdictions in real time, akin to software updates, thereby achieving future-proof resilience.

1. Modular Compliance Engine

We decoupled the compliance logic from the core transaction logic and encapsulated it as a

standalone plug-and-play module.

- Hot-Swappable architecture:
    - The core OES guarantee contract (responsible for locking and releasing funds) remains unchanged, ensuring the immutability of the underlying layer.
    - The outermost Compliance Hook can be upgraded at any time based on governance voting.
    - Scenario: If the EU abruptly lowers the anti-money laundering threshold from 1,000 to €0, DAO would only need to vote to update the EU_AML_Module parameters, without requiring protocol migration or mandatory user withdrawals.

2. Jurisdiction-Specific Adapters

The OmniPact interface dynamically loads different compliance interfaces based on the user's digital footprint (IP address and wallet history).

- US Mode:
    - Automatically hide high-risk RWA (Regulatory Waiver Agreement) yield products to comply with securities laws.
    - Mandatory implementation of enhanced KYC verification standards.
- Global Mode:
    - Provide full functionality for regions with less stringent regulatory requirements.
    - Only basic AML blacklists are blocked.
- The rationale: This tailored compliance strategy enables us to safeguard our core market (Web3 Degen) while securely serving the compliance market (institutions).

3. DAO Compliance Committee

To proactively address legal risks, DAO will establish a dedicated Legal Guild.

- Team Composition: Engage top-tier external crypto law firms (e.g., Latham & Watkins, DLA Piper) as advisory partners.
- duty：
    - Monitor the legislative developments in key global markets on a continuous basis.
    - Draft an OCR (Open Compliance Request) proposal recommending the DAO to amend its protocol parameters to comply with the new regulations.
    - Example: When a privacy coin is removed from multiple countries 'platforms, the committee may propose its removal from OmniPact supported list.

4. Circuit Breaker & Isolation

The protocol demonstrates emergency survivability when confronted with extreme regulatory black swan events, such as a government's abrupt ban on DeFi.

- Regional outage: Allows service suspension for specific IP addresses or cryptocurrencies (e.g., algorithmic stablecoins) in designated regions, without disrupting normal operations for global users.
- Front-end Decentralization:
    - Even if the official omnipact.com is shut down by regulators, because the protocol is deployed on IPFS and the chain, the community can quickly deploy multiple unofficial frontends (Mirrors) to ensure that users' assets can always be

extracted and eliminate the risk of "shutting down and running away".

9. Risk Management and Security System

As a global protocol facilitating high-value asset transfers, OmniPact prioritizes security as its core principle. We have moved beyond the outdated single-point defense model to establish a comprehensive Defense in Depth framework that integrates technical safeguards, operational risk management, and market dynamics.

9.1 Comprehensive Risk Prevention System

To address potential risks in decentralized collateralized trading—including fraudulent reporting, code vulnerabilities, malicious arbitrage, and compliance issues—we have implemented a rigorous end-to-end governance framework covering pre-transaction, in-process, and post-transaction phases.

Phase 1: Pre-Event Prevention-Building a Robust Access Control and Code Protection System

Core Strategy: By enforcing high-standard thresholds and rigorous code audits, we effectively block 90% of risks from protocol execution.

1. Smart Contract Security

- Multi-round cross-audits: The core OES contract code must undergo independent audits by at least three top-tier security agencies (e.g., CertiK, Trail of Bits, OpenZeppelin) prior to mainnet deployment, with all audit findings and remediation records publicly disclosed.
- Formal Verification: This method employs mathematical modeling to validate critical systems like fund pool management (Vault) and access control (Access Control), ensuring theoretical proof that the code will never encounter logical deadlocks or overflow issues under any operational conditions.
- Parameterization restrictions: Set hard-coded "security boundaries" at the contract level. For example, the maximum locked amount (Cap) for a single contract to prevent massive losses caused by early-stage code vulnerabilities.

2. Anti-Fraud Submission

- Evidence standardization: Sellers must submit evidence in a standardized format (e.g., with timestamps and geographic metadata). For image evidence, AI algorithms perform pre-scanning to automatically identify and reject PS-modified fake images.
- Reputation Gating: For high-risk categories (e.g., large unsecured loans), a minimum Omni-Score is required to access. Users with low credit scores cannot initiate or participate in such high-risk transactions.

3. Compliance Isolation

- Front-end geo-fencing: Automatically blocks access requests from sanctioned regions (OFAC-listed) or high-risk jurisdictions through IP identification and wallet behavior analysis.

- Pre-Blacklist Check: Integrates compliance oracles like Chainalysis to perform AML scans before funds enter the protocol. Transactions from suspicious sources (e.g., mints) will be rejected.

Phase 2: In-Event Monitoring-The Eagle Eye System for Real-Time Perception

Core Strategy: Utilize automated probes and AI algorithms to detect and block anomalies in real time.

1. Market Arbitrage and Manipulation Protection

- Multi-source verification for oracles: For transactions dependent on external prices (e.g., insurance claims), Chainlink DON multi-node pricing is implemented with outlier removal to prevent price manipulation through lightning loan attacks.
- MEV Protection: During cross-chain settlement, integrate Flashbots Protect or similar private RPC nodes to prevent trading bots from executing sandwich attacks in the Mempool, which exploit users' slippage.

2. Transaction Circuit Breaker

- Traffic risk control: The system monitors the real-time outflow rate of the total locked volume (TVL) across all protocols.
    - Trigger conditions: If abnormal capital outflows surge within a short period (e.g., 50% of TVL withdrawn in 1 hour), or if there are multiple small withdrawals from the same IP (suspected of a dust attack).
    - Response: The smart contract automatically activates "Pause Mode", freezing all non-urgent withdrawal functions until the security committee investigates and resolves the issue.

3. User Asset Security Guard (Asset Sentinel)

- Abnormal interaction alert: The DApp has built-in security plugins. A high-risk red alert will pop up if a user attempts to approve unlimited quotas for an unknown new contract or if the counterparty address is flagged as fraudulent.

Phase 3: Post-Event Remediation — Comprehensive Compensation and Legal Recourse

Core Strategy: When defensive measures fail, minimize user losses through financial compensation and legal remedies.

1. Insurance Fund Compensation

- SAFU Fund: 10% of the revenue from the agreement treasury is automatically transferred to SAFU (User Asset Security Fund).
- Scope: In cases where funds are stolen by hackers due to inherent vulnerabilities in smart contracts (not user errors) or where oracle malfunctions cause misjudgments, the DAO will initiate a SAFU fund vote to provide full or partial compensation to affected users.

2. Arbitration and Penalty (Arbitration & Slashing)

- Punishment for misconduct: If a party is found to have falsified transactions (e.g., fraudulent shipments), they must compensate the buyer for losses and forfeit the full amount of their pledged margin (Slash).
- Reputation Deactivation: Accounts with Omni-SBT misconduct will be flagged as 'Fraudsters'. This permanent record on the blockchain will permanently block them from

generating commercial profits in OmniPact and its partner ecosystem.

3. Legal Recourse Support

- Evidence Export: OmniPact provides a 'One-Click Judicial Evidence Package Export' feature for RWA or large-scale commercial fraud cases. The package includes digitally signed contracts, fund flow records, and arbitration awards, meeting the electronic evidence standards of most common law jurisdictions, enabling users to file lawsuits in offline courts.

9.2 Emergency Response Plan: Anti-Fragility Mechanisms

OmniPact operates under the security philosophy of "pessimistic assumption, optimistic execution." We assume the system could be attacked at any moment, hence establishing a 24/7 emergency response protocol (ERP). This mechanism ensures user capital remains secure even during extreme "black swan" events, such as public chain outages, stablecoin de-anchoring, or zero-day vulnerability attacks.

9.2.1 Organizational Structure: Security Council & War Room

To tackle the inefficiency in DAO voting, we've activated a' wartime mode' during the emergency.

- Security Guardian Council:
  - The entity is composed of 9 core technical members, representatives of major investors and external security experts.
  - Permissions: You have the highest permission to trigger "Emergency Pause" and "Protocol Upgrade".
  - Response time: Commit to 24/7 availability, ensuring a response within 15 minutes of a security incident being detected.
- Virtual War Room:
  - Upon triggering the highest-level alert, immediately activate the encrypted meeting channel to assemble the development, legal, and PR teams, along with collaborating security audit firms like CertiK, for coordinated action.

9.2.2 Smart Contract Vulnerability Response SOP (Vulnerability Response SOP)

For critical code-level vulnerabilities (such as the risk of fund theft), we follow these standard procedures:

1. Detect & Intercept:
   - The on-chain monitoring probe (Sentinel) detected abnormal capital outflows.
   - Pause Guardian: An automated script or committee member immediately calls the pause() function.
   - Result: All deposit, withdrawal and transaction functions of the protocol are frozen instantly. Even if the attacker finds the vulnerability, he cannot withdraw the funds.
2. Whitehat Rescue:
   - If funds are at risk of theft but not yet transferred, the committee may authorize a 'white hat operation' to exploit the same vulnerability and preemptively move the remaining funds to a secure cold wallet (Multisig Treasury) for protection.
3. Fix & Upgrade:

    - The development team replicated the vulnerability on the test network and created a patch.
    - Security audit companies conduct emergency audits.
    - Deploy new contracts via the Timelock accelerated channel.
4. Resume & Compensate:
    - System thaw. The insurance fund is activated to compensate affected users, and a detailed Post-Mortem (post-mortem report) is published.

9.2.3 Economic Black Swan Response

We have a dedicated circuit breaker mechanism to handle external market crashes (e.g., the LUNA zeroing-out incident) or infrastructure failures.

- Stablecoin Depeg Protection:
    - Scenario: Suppose the settlement currency USDC, supported by the protocol, suddenly experiences a severe de-anchoring (e.g., dropping to $0.8).
    - Response: Oracle detects abnormal price deviation. The protocol automatically suspends the asset's pricing and settlement functions to prevent users from exploiting price spreads for malicious arbitrage or low-price settlements.
- Oracle Failure Circuit Breaker:
    - Scenario: The Chainlink or API data source returns incorrect prices (e.g., ETH = $1).
    - Response: When the oracle quote fluctuates beyond the threshold (e.g., ±20%) or multi-source data is inconsistent, the contract automatically rejects transactions dependent on this price until manual intervention confirms.

9.2.4 Response to Extreme Force Majeure (Force Majeure Response)

Extreme crackdowns on the physical world or regulatory level.

- Frontend Censorship Resistance:
    - Scenario: The official domain name omnipact.com is blocked by an ISP in some countries or confiscated by a domain name registrar.
    - Response: Enable IPFS + ENS decentralized front-end. The community can access the protocol directly through omnipact.eth or IPFS hash, and no centralized force can cut off the service.
- Public Chain Halted:
    - Scenario: A full network outage occurs on a decentralized public blockchain (e.g., Solana).
    - Response: OmniPact State Keepers automatically record the latest valid block state. Once the public chain recovers, the system will perform state synchronization based on timestamps and grant affected users an extended' dispute appeal window'.

9.3 Potential Risk Disclosure

The OmniPact team works to minimize risks through rigorous code audits, economic model stress testing, and compliance architecture design. However, given the high volatility of the

cryptocurrency market and the experimental nature of blockchain technology, participating in the OmniPact ecosystem—whether through collateral services or holding $PACT tokens—still carries significant potential risks.

Before making any decision, users should fully understand and assess the following risk categories:

1. Technical and Security Risks

Despite our implementation of a defense-in-depth system, the inherent nature of software engineering still leaves the following possibilities:

- Smart contract vulnerabilities: Despite multiple rounds of audits, complex logic—particularly in cross-chain and arbitration components—may still contain undetected zero-day exploits, potentially leading to frozen or lost funds.
- Underlying public chain failures: OmniPact depends on the stability of core public chains like Ethereum and BNB Chain. If these chains experience congestion, forks, rollbacks, or 51% attack scenarios, the protocol's normal operation may be affected by force majeure.
- Prophecy manipulator risk: The RWA business relies on external data sources (e.g., logistics API). If the data source is hacked or prophecy nodes collude to feed false data, it may result in incorrect fund releases.

2. Market and Economic Risks

- Token price volatility: The $PACT token's value is subject to global macroeconomic conditions, regulatory policies, and market sentiment, which may cause sharp fluctuations in a short period and even risk a zero value.
- Stablecoin de-anchoring risk: OmniPact collateral is primarily denominated in stablecoins like USDT/USDC. Should the issuer (Tether/Circle) face a repayment crisis causing de-anchoring, users' principal value would be compromised.
- Liquidity risk: In extreme market conditions, decentralized exchanges (DEX) may face liquidity shortages, preventing users from buying or selling $PACT tokens promptly.

3. Compliance and Regulatory Risks

- Legal landscape evolution: Regulatory frameworks for DeFi and DAO remain ambiguous and subject to dynamic changes across jurisdictions. In the future, certain countries may enact new laws to restrict or ban access to OmniPact, or classify $PACT as illegal securities.
- Tax liability: Users are responsible for declaring and paying capital gains tax or VAT on OmniPact usage based on their tax resident status. The agreement does not assume withholding obligations.

4. Operational and Governance Risks

- Governance attack: If a malicious entity (or consortium) controls over 51% of $vePACT voting rights, they could alter protocol parameters via DAO proposals or attempt to siphon off treasury funds. While time locks and guardian veto mechanisms provide some protection, the risk remains theoretically present.
- Conspiracy of arbitrators: Although there is the Schelling point game mechanism, in extreme cases, if most arbitrators are controlled or bribed by the same interest group, it

may lead to unfair decisions.

5. Other Potential Risks

- Quantum computing poses a threat: future mature quantum computing technology may crack existing asymmetric encryption algorithms (such as ECDSA), jeopardizing the security of private keys for all blockchain users.
- Internet infrastructure failure: Large-scale global internet outages or cloud service provider (AWS/Google Cloud) outages may temporarily make front-end services unavailable.

Disclaimer: This white paper contains no investment advice, legal counsel, or purchase offer. OmniPact is provided 'as is' (AS IS), and the development team shall not be liable for any direct or indirect losses arising from its use. Users are solely responsible for managing their private keys and safeguarding their assets.

Please visit www.omnipact.io to view the relevant terms and conditions.